

A formal knowledge base for amplicon sequencing data and geochemistry of the Rub' al Khali desert

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Abstract

Integrating multi-omic data with environmental metadata remains a significant challenge due to the heterogeneity of data formats and a lack of standardized semantic representations. Building such an integrated resource requires addressing several interlocking challenges: defining a unified data model across instruments and observation types, harmonizing taxonomic identifiers across reference databases, validating incoming data against shared schemas, We collected soil and supporting semantic query answering at scale under a tractable reasoning regime. Here, we present a high-resolution microbiome dataset from plant material during five expeditions campaigns across a 1,043 km transect of the Empty Quarter (Saudi Rub' al-Khali al-Khali (Empty Quarter)) of Saudi Arabia, comprising 2,516 sample individuals from 60 primary sampling sites and special-interest locations. A The resource links 2,550-row source ledger connects specimens, DNA extractions, libraries, sequencing runs 516 non-control specimen entries with laboratory processing and community profiles; the a canonical feature amplicon table contains of 1,271 taxonomic profiles, of which 1,237 pass ecological filtering control and read-depth exclusions. These biological sequences are joined to Associated observations comprise 274 field environmental logs, 12,936 monthly climate measurements, 712 archived-soil pH observations measurements, and two X-ray fluorescence workflows (71 field X-ray fluorescence sessions and 725 laboratory X-ray fluorescence records) and 3,234 site-month climate summaries. We augment these data with represent sites, specimens, processes and measurement values in a formal modular RDF knowledge base constructed using OWL 2 DL and the Semanticscience Integrated Ontology (SIO) and validated chemical, with ShEx-based ingest validation environmental and ELK-based consistency checking over the resulting graph taxonomic references. This semantic layer enables complex cross-domain queries Shared identifiers preserve specimen identity and provides assay-specific provenance. Validation checks source reconciliation, measurement admission, identifier mappings, structural constraints and query outputs. The graph supports retrieval across these linked observations through a comparative baseline for studying microbial adaptation in extreme environments web portal and SPARQL endpoint. Versioned observations, graph modules and workflow code support comparisons across sites, campaigns and assay types.

Keywords: Knowledge Graph graph, Metagenomics Amplicon sequencing, Rub' al Khali, SIO, Semantic Web web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is the largest contiguous sand desert on Earth, a hyper-arid environment that spans approximately 500,000 km² of the southern Arabian Peninsula¹. The region is characterized by extreme atmospheric temperatures that fluctuate, negligible and highly Its soils experience large temperature fluctuations and irregular rainfall², and high humidity in the western edge³. Despite these abiotic stresses, hyper-arid desert soils harbor yet support diverse microbial communities⁴. To predict ecosystem responses to climate change and to inform conservation and

land-use policy, researchers require deep knowledge of the biogeochemistry of such extreme environments ⁵.

Saudi Arabia already has a substantial arid soil microbiome literature. Previous studies Repeated observations across soil compartments and seasons provide a basis for studying these communities together with their environmental conditions.

Previous Saudi Arabian surveys examined elevation gradients³, rainfall responses across soil compartments⁶, and bulk and rhizosphere soils across ~~three regions~~⁷, rhizosphere bacteria of pioneer sand plants near Jizan⁸, and bacteria associated with native desert plants⁹. More recent surveys characterized fertility islands and plant-associated networks^{10,11}, sandy soils near natural and artificial water points¹², rhizosphere and endosphere communities across northern reserves¹³, and Red Sea coastal sabkhas¹⁴. Fungal studies progressed from a small number of bulk soil samples from the western coast^{15,16} to a standardized single survey of 85 locations across Saudi Arabia¹⁷, and a recent synthesis identifies substantial geographic and taxonomic gaps in Arabian biological soil crust research¹⁸. In the United Arab Emirates, related studies profiled a Rub' al-Khali interdune microbial mat¹⁹ and desert-plant rhizosphere and endosphere communities²⁰. These studies establish that a survey of hyperarid soils in Saudi Arabia is not itself new. What remains missing is a repeated, multi-expedition transect in which every community profile the sample it came from, and the measurements taken alongside it form a connected record that can be queried directly rather than assembled from different data records.

To characterize complex ecosystems, researchers must integrate heterogeneous and multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that link physical samples to their digital representations. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility.

Engineers originally developed the concept of a digital twin, and environmental scientists increasingly adopt this approach^{21,22}. Recent frameworks, such as TwinEco, standardize the development of dynamic, data-driven digital twins in ecology to avoid fragmented software designs²³. Additionally, regional natural resource management benefits from artificial intelligence-assisted digital twin architectures that unify heterogeneous environmental observations²⁴. Such digital twins show strong predictive accuracy when they incorporate continuous data streams, such as citizen science observations for real-time biodiversity forecasts²⁵. To guarantee that these models remain interoperable and reusable across domains, researchers emphasize the integration of datasets, models, and computational workflows through Findable, Accessible, Interoperable, and Reusable (FAIR) principles²⁶. However, while biological systems and specific regions have digital representations²⁷, researchers have not yet developed semantic representations of entire desert ecosystems. Furthermore, ecosystem digital twins largely focus on macroscopic scale and do not cover different levels of granularity; modeling entire landscapes and environments, but not microbial communities.

Constructing such a resource for an entire desert ecosystem requires addressing several interlocking challenges that have shaped the design of this work. First, the data are produced by heterogeneous instruments and protocols (16S amplicon sequencing, X-ray fluorescence, GPS-tagged field notes, gridded climate records), so a *unified data model* is needed that can represent samples, processes, qualities, and quantities under a common upper-level vocabulary rather than as disconnected tables. Second, taxonomic labels are reported by multiple classifier workflows as text strings, and the same label can denote different taxa across reference databases and releases, so they must be harmonized against a single reference taxonomy without losing the original assertions. Third, because data are deposited in batches over five field expeditions, *schema validation at ingest* is required to catch structural errors before they propagate into the graph. Fourth, the knowledge graph must support *semantic query answering at scale*: with tens of millions of triples, expressive OWL 2 DL reasoning is computationally infeasible, so a tractable reasoning regime must be selected and its trade-offs made explicit. Fifth, a *user-facing query and exploration interface* is required so that domain scientists who are not fluent in SPARQL can interact with the resource. We return to each of these challenges in the usage notes (Section 5) and link them to the corresponding design choices in the methods and knowledge representation sections.

[regions](#) ⁷. [Studies of desert plants further connected microbial profiles with host-associated habitats](#) ⁹. A repeated transect adds a specific data-management requirement: each community profile must remain connected to its field specimen, collection visit, laboratory processing and associated measurements. [Site identity alone is insufficient for temporal comparisons because a site can have several visits and assays can use different field replicates.](#)

Community reporting standards and knowledge-graph infrastructure define the reuse target this resource is built against. The FAIR principles set the criteria a deposited resource is judged against ²⁸[Multi-campaign resources such as Tara Oceans link campaign, station and sampling-event registries to environmental and sequence data](#) ²⁹. [MIxS/MIMARKS](#) [define specify](#) contextual fields for [marker-gene and environmental sequence data](#) ³⁰, and [the STREAMS guideline extends reporting practice to](#) [STREAMS extends reporting guidance for](#) environmental microbiomes ³¹. The National Microbiome Data Collaborative [shows how persistent identifiers, standardized metadata and linked-data schemas support reuse across environmental microbiome studies](#) ³². On the knowledge-graph side, [uses persistent identifiers and shared schemas to connect environmental microbiome data](#) ³². [Knowledge graphs such as KG-Microbe connects microbial taxa with traits, environments and growth preferences](#) ³³, and [MetagenomicKG integrates](#) ³³ and [MetagenomicKG](#) ³⁴ [organize taxonomic, functional and biomedical sources for metagenomic applications](#) ³⁴. These resources establish that a microbiome knowledge graph is not itself novel. The contribution of the present resource is the joined, expedition-level provenance of locations, physical specimens, laboratory processes, geochemistry, climate, sequence runs, quality control and taxon observations from [environmental information across sources](#). We apply these principles to the specimen and process provenance of a single landscape survey, rather than vocabulary coverage or graph construction alone.

Here, we describe a versioned, provenance-preserving knowledge graph integrating repeated desert microbiome, environmental, climate, pH, and geochemical observations across the Rub’ al-Khali. We present an OWL 2 DL knowledge graph ³⁵ built from RDF instance data ³⁶. This knowledge graph integrates data from five field expeditions. We collected soil and plant material during five campaigns (2023–2025) along a transect of more than 1,000 km ^{043 km transect} across the Saudi Rub’ al-Khali, from the western escarpment to the eastern border with Oman. A versioned source ledger of . The catalogue contains 60 primary sites and 10 named locations. A 2,550 rows links field specimens to their downstream products; 550-row specimen catalogue contains 2,516 non-control rows are represented as sample individuals in the graph. An amplicon feature specimen rows. The canonical amplicon table contains 1,271 profiles, of which; the companion ecology analysis retains 1,237 pass ecological filtering across 64 numeric site labels. We incorporated X-ray fluorescence (XRF) measurements from two complementary workflows: in-situ handheld field sessions , and processed laboratory records of archived material from all five expeditions. We further integrated site-specific environmental metadata recorded at each visit (274 field logs), 712 admitted after control and read-depth exclusions. The resource links these profiles to field environmental observations, archived-soil pH observations, and monthly climate records derived from the Open-Meteo database ³⁷, field and laboratory XRF measurements, and climate data. The coverage tables distinguish specimens, profiles and assay observations.

We structured the knowledge graph encoded sites, specimens, processes, qualities and measurement values with the Semantic Science Integrated Ontology (SIO) ³⁸ as the upper-level framework. This design allows interoperability with other SIO-based datasets. We modeled domain-specific concepts as subclasses of SIO entities, and all measurements follow SIO’s ontology design pattern for measurement processes. We grounded chemical entities in the Chemical Entities of Biological Interest (ChEBI) ontology ³⁹, environmental contexts in the Environment Ontology (ENVO) ⁴⁰, and taxonomic identifiers in the ChEBI ³⁹, ENVO ⁴⁰ and NCBI Taxonomy ⁴¹ provide chemical cross-references, environmental classes and validated taxonomic identifiers. Classifier lineages retain their provenance alongside validated identifiers and explicit unresolved mappings. The resulting RDF modules support queries across these linked observations.

Because it encodes the full provenance chain from physical sample collection to digital data products, the knowledge graph is the key foundation for implementing the FAIR data principles for the Rub’ al-Khali digital twin. This semantic integration eliminates the data fragmentation that typically limits ecological models, and it guarantees that observations, models, and workflows remain fully interoperable ²⁶. Consequently, the knowledge graph acts as the operational foundation for the Rub’ al-Khali digital twin. It allows researchers to execute cross-domain queries that otherwise demand complex joins across multiple heterogeneous tables in a conventional database. Therefore, this resource serves as a foundational infrastructure for computational ecology, which facilitates hypothesis generation, intervention simulation, and evidence-based policy. Technical validation examines source reconciliation, measurement admission, identifier mappings, structural

constraints and executable query examples. We report the scope of each check separately, including the constructs and modules processed by ELK.

~~This resource makes several contributions. On the scientific side, it provides a coordinated multi-trip desert microbiome and geochemistry dataset—16S rRNA amplicon profiles, XRF elemental measurements, archived soil pH, GPS-tagged field metadata, and monthly climate records for 2,516 sample individuals from five expeditions across a 1,043 km transect of the Rub’ al-Khali, together with a harmonized taxonomic abundance table that maps amplicon sequence variants onto NCBI Taxonomy while preserving the original assertions. On the technical side, the resource comprises an OWL 2 DL knowledge graph built on SIO with domain classes grounded in ChEBI, ENVO, and NCBI Taxonomy, modelling the full provenance chain from field sampling to digital data products. We formalize reusable ontology design patterns for sampling, measurement, taxonomic abundance, and provenance aligned with SIO’s core idioms. A ShEx-based ingest validation workflow enforces structural shapes on every RDF batch before loading, and the ELK reasoner is used to validate logical consistency of the knowledge graph. A set of competency questions documents the intended query workload and serves as regression tests, and a web portal with an interactive map, per-site dashboards, and a SPARQL editor exposes the knowledge graph to domain scientists without requiring SPARQL fluency.~~The portal at <https://rubalkhali.science/> exposes the graph through site pages and a SPARQL endpoint. Data Records describes the distributed datasets and Data Availability gives their access conditions. The companion ecological analysis reports the inferential results for the quality-controlled community profiles, laboratory-XRF gradient, rainfall, spatial turnover and association networks⁴²; the present paper describes the reusable data products, provenance model, identifier mappings and their executable validation. ecology manuscript reports the community, environmental-association and functional analyses⁴²; this Data Descriptor documents their source data, representation and reuse conditions.

2 Methods

2.1 Study area and sampling design

We conducted five sampling ~~expeditions campaigns~~ across the Rub’ al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E). The 60 ordered sampling sites yield a 1,043-km cumulative path and lie 1,015 km apart end to end. Figure 1 ~~illustrates the altitude profile across this transect~~shows the site coordinates and altitude profile. Cumulative distance sums great-circle distances between successive sites, using an Earth radius of 6,371 km.

The transect comprises 60 primary sampling sites at predetermined ~~geographic coordinates~~, coordinates and 10 named ~~sites of special interest (water locations, including wells, a hot spring, and an oil spill site)~~and an oil-spill site. Trip 1 ~~additionally used the numeric source identifiers 61–64.~~ Matching their coordinates against the field records shows that these refer to four of the sites of special interest rather than further locations, so we retain the numeric identifiers as source provenance

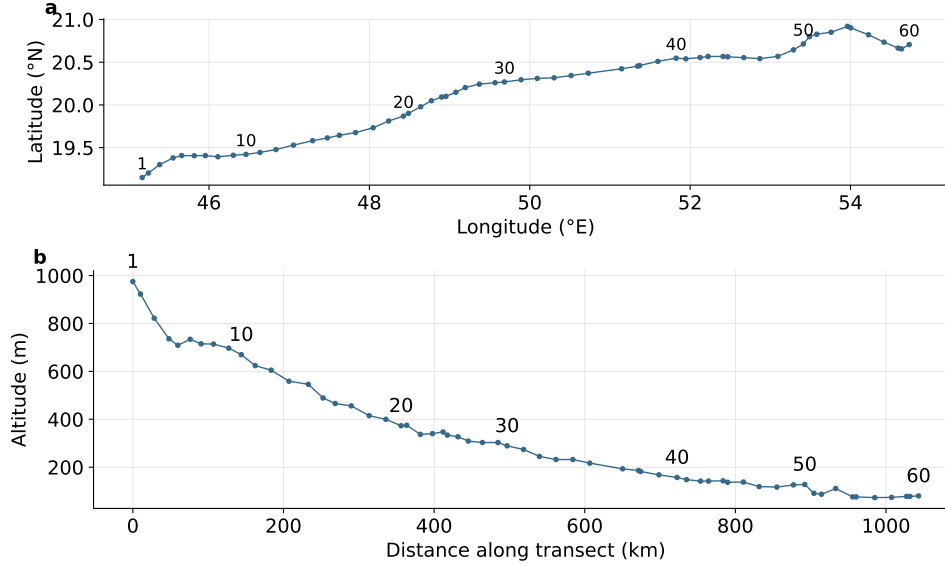


Fig. 1 Altitude profile along the Primary sampling transect across the Rub' al-Khali. Panel a shows the coordinates of sites 1–60; panel b shows their archived altitude values against cumulative great-circle distance in site order. Labels identify selected sites. Coordinates and altitude come from the western escarpment to versioned site-altitude table; the eastern border with Oman elevation source and vertical datum remain undocumented.

but do not count them as four additional locations. Because these locations were not revisited on later trips, we exclude their records from analyses on the repeated campaign of are aliases for four named catalogue locations, established by coordinate matching, and account for 36 source specimens. The repeated-campaign analytical frame uses sites 1–60. The metadata link each observation to its site, campaign and specimen.

To capture temporal variability, we conducted expeditions campaigns across different seasons: late winter 2023 (Trip 1, 64 sites visited), summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures), winter 2024 (Trip 3, 65 sites visited), summer 2024 (Trip 4, 60 sites visited), and autumn 2025 (Trip 5, 62 sites visited). We recorded Field observations provide temperature, atmospheric pressure, and relative humidity at each visited site in each expedition with campaign-specific availability. We summarize the observational coverage in Table 1. We acquired X-ray fluorescence (XRF) data through two separate workflows that are not interchangeable replicates: handheld in-situ We acquired handheld field XRF measurements during Trip 5, and laboratory and laboratory XRF measurements of archived material from all five expeditions. We compiled an environmental table from the original field sheets, recording every correction individually rather than editing the sheets, and retained all campaigns. The curated environmental table retains 274 non-blank records, nonblank records, nonblank source rows. Of these, 260 resolve to catalogue site visits and generate 260 temperature,

187 pressure and 251 humidity values. Fourteen auxiliary or revisit labels remain table-only pending unambiguous site identification. Trip 2 includes eight temperature measurements spanning 34.5–41.9°C; pressure and humidity were unavailable for that campaign. Field observations entered the dataset only when their site identity and measurement range could be established.

Table 1 Coverage of field environmental logs. Numeric identifiers, named locations and auxiliary observations retain separate source labels.

Campaign	Season	Field-log records	Measurements available
Trip 1	Late Winter 2023	1–60; 4 named; 15 auxiliary*	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

*The 15 observations dated March 2023 represent Trip 1 auxiliary/revisit observations.

~~At each site, we collected samples from three distinct~~ The sampling design targeted ~~three~~ soil compartments: bulk surface soil (~~0 cm depth~~), bulk subsurface soil (nominal 5–10 cm depth, hereafter “deep”), and subsurface soil collected near roots (5–10 cm depth, hereafter “root-adjacent”). ~~We obtained~~ The ontology class Deep Soil Sample denotes the 5–10 cm bulk compartment, called shallow-subsurface in the companion ecology analysis⁴². We collected three co-located field replicates per compartment ~~by placing soil into sterile tubes, either by scooping soil directly into the tube or using~~ using sterile tubes and sterile plastic spoons; ~~these replicates are not independent sites~~. From Trip 2 onwards, three adjacent tubes were taped or chained together for collection, so adjacent sampling positions were separated by one external tube diameter. Trip 5 used larger tubes. Exact tube diameters, surface sampling depth, individual bulk subsurface depths and distance from roots were unrecorded. These co-located replicates share a site and compartment. We preserved samples at –20°C in the field using an Engel MR040F portable freezer. At the end of each ~~expedition~~ campaign, we transferred samples to an –80°C freezer in the laboratory.

~~We included positive and negative controls for each sampling campaign to mitigate and control for contamination artifacts. The positive control design differed by expedition~~ The specimen catalogue additionally contains 248 plant entries linked by campaign, site and source identifier. Tissue type, host-identification procedure, vouchers and plant-specific storage details are incomplete in the archived protocol. These entries preserve plant provenance; root-adjacent soil retains its own specimen identity.

2.2 Laboratory controls

Control coverage differed by campaign and processing stage (Table 3). Extraction blanks underwent the same procedure as biological specimens. Trips 4–5 used one

Table 2 Source-specimen coverage by campaign and soil compartment, linked through the specimen catalogue. All catalogue sites are included; plant and control rows are excluded. Columns count source rows eligible for specimen representation, expected by the extract generator, linked to at least one run, present in the ASV table and retained in the ecological analysis. Counts apply to source rows with a profile link; sequencing-profile totals are reported separately, and repeated profiles can share a specimen row.

Trip	Compartment	Specimens	Extract-linked	Run-linked	ASV table	Ecology
1	Subsurface	192	184	111	110	108
1	Root-adjacent	192	170	102	106	104
1	Surface	192	185	108	114	113
2	Subsurface	24	23	9	8	7
2	Root-adjacent	24	18	13	13	11
2	Surface	24	22	7	6	6
3	Subsurface	180	180	141	141	141
3	Root-adjacent	180	174	154	154	154
3	Surface	180	180	156	155	155
4	Subsurface	180	63	59	59	59
4	Root-adjacent	180	60	58	58	58
4	Surface	180	62	60	60	60
5	Subsurface	180	93	57	55	55
5	Root-adjacent	180	151	133	133	133
5	Surface	180	82	30	29	29

blank per trip-specific extraction-day batch. Trips 1–3 used one blank per extraction kit, and Trips 1–2 included shared extraction days. The PowerSoil Pro control has a sequencing record ([Extraction-Ctrl-Pro-Trip1](#), library M-25-0929); the PowerSoil blank is present in laboratory records, with no matching sequencing record located in the available records.

The canonical table contains 18 Trip 5 extraction-blank profiles (EB1–EB18) and six no-template PCR controls (Negative1, Negative2, Negative4–Negative7). Seventeen extraction blanks have confirmed batch links to 217 biological profiles. The six Trip 4 extraction blanks (SEB, EB1–EB5) were sequenced on a shared run and denoised separately with the same pipeline on 30 August 2026. Three of the nine Trip 4 extraction batches had unsequenced blanks, leaving complete or partial coverage gaps at 28 sites. Batch assignments identify individual specimens; the profile inventory excludes unsequenced S34Dr1 and S40PRr1. S40PRr1 failed three amplification attempts. S34Dr1 yielded DNA but was probably missed during pooling.

No-template controls contained PCR reagents and molecular-grade water in place of template DNA. Reagent-only PCR blanks contained neither added water nor template. One reagent-only blank was prepared per amplification plate. Most showed low or undetectable concentration and adapter-dimer-dominated TapeStation profiles and were excluded from sequencing. Trip 5 reagent-only blanks were lost during handling, attributed to evaporation. Trip 4’s PCR blank and NTC were excluded because their index pairs conflicted with biological libraries on the shared run. The Trips 1–3 control libraries were sequenced on NovaSeq 6000 A01018 run 321, flowcell

Table 3 Control coverage relevant to the released amplicon data. Detailed material and sequence occurrences are retained in the control registry.

Campaign	Extraction controls	PCR-stage negatives	Sequenced standards
1	Per-kit; Pro blank sequenced	Nuclease-free water	Identity unresolved
2	Shared per-kit cover- age	NTC-2	Two purified-DNA standards
3	Per-kit coverage	PCRCtrl, index 728	Two whole-cell stan- dards
4	Six sequenced blanks; three unsequenced batches	PCR blank and NTC unsequenced	Amplification-positive control only
5	18 canonical blank pro- files; 17 batch-linked	Six canonical NTC profiles	Field standard sequenced by shotgun

[HVJY5DRX3](#), and denoised with biological libraries in July 2025; they are distributed separately from the canonical feature table. Re-denoising on 30 August 2026 supplied the control comparison in Technical Validation.

Positive standards were reserved for genus-recovery evaluation. Trips ~~1 and 2~~ [1-2](#) used the ZymoBIOMICS HMW DNA Standard D6322, ~~supplied as purified high-molecular-weight DNA, which we extracted along with the samples after the trip, introduced at PCR;~~ Trip 3 used the whole-cell ~~ZymoBIOMICS D6300~~ [Microbial Community Standard D6300](#), ~~which we also later extracted, which also undergoes extraction.~~ Trip 4 had ~~no positive control, a laboratory amplification-positive control assessed by gel electrophoresis and TapeStation, and~~ Trip 5 ~~used~~ [its field](#) D6300, ~~but its positive and negative control pair were standard was~~ sequenced by shotgun metagenomics ~~rather than by the 16S assay.~~ [Four sequenced standards from Trips 2-3 support the amplicon genus-recovery analysis. The Trip 1 FPosCtrl1 identity remains unresolved because its index and well disagree, so it remains excluded.](#)

D6322 contains seven bacterial genomes at 14% genomic DNA each and ~~*Saccharomyces cerevisiae* one yeast~~ [one yeast](#) at 2%; D6300 contains eight bacterial genomes at 12% each and two yeasts at 2% each. These ~~are~~ [manufacturer](#) genomic-DNA ~~fractions are composition specifications, not expected 16S read fractions.~~

~~Negative controls included distinct extraction blanks and PCR blanks. One extraction blank was included per extraction day, and an extraction day could contain samples from several expeditions, so an extraction blank is linked to its extraction batch and not directly to a trip. Trip 4 records five numbered blanks and one unnumbered blank, but the blank covering a three-day extraction was not sequenced, leaving all or part of 23 sites without a sequenced blank. Author-confirmed controls are recorded in a ground-truth table; identity, batch-membership and assay-applicability claims that await laboratory confirmation are represented in the graph as evidence-bearing dispositions rather than as ground truth, and no field-blank instance is asserted because no sterile-bag inventory was recovered~~ [mass fractions. The recovery analysis compares expected and observed bacterial genera,](#)

including unassigned reads and additional taxa. Such departures can reflect processing contamination, amplification, classification or index-assignment errors. The registry retains intended materials, observed profiles and identity evidence separately.

2.3 DNA extraction and sequencing

~~We utilized Genomic DNA was extracted from 0.9–1.2 g of soil for DNA extraction using the~~ using either the QIAGEN PowerLyzer PowerSoil Kit (QIAGEN, catalog #catalogue no. 12855-50) or the QIAGEN DNeasy PowerSoil Pro Kit (QIAGEN, catalog #catalogue no. 47014). ~~We homogenized all samples with the~~ Both kits were used in Trips 1–3: among the sequenced profiles, the specimen metadata identify the PowerLyzer PowerSoil Kit for 77 Trip 1, 10 Trip 2 and 359 Trip 3 profiles (54 of the Trip 3 records name the MoBio PowerSoil kit) and the DNeasy PowerSoil Pro Kit for 217 Trip 1, 11 Trip 2 and 87 Trip 3 profiles; 46 Trips 1–3 profiles have no recorded kit. The QIAcube Connect was introduced for routine extraction starting with Trip 4; samples from Trips 1–3 were also processed using the QIAcube Connect when re-extraction was required. The extraction method and re-extraction history were recorded for each specimen when available.

Frozen soil samples were thawed on ice and processed immediately after thawing. Before homogenization, samples were incubated at 65°C for 10 min with agitation at 1,200 rpm, then homogenized using a PowerLyzer 24 Homogenizer ~~and completed extraction at 4,000 rpm for one 45-second cycle.~~ The remaining purification steps were performed either manually or ~~via the automated QIAcube Connect system.~~ ~~We quantified the eluted DNA using a QIAcube Connect instrument, following the manufacturer’s protocol.~~ DNA was eluted in 50 µL of the elution buffer supplied with the respective kit.

Extraction controls followed the campaign-specific scheme above.

Extracted DNA was quantified using the Qubit ~~HS dsDNA kit.~~

~~We amplified bacterial DNA dsDNA~~ High Sensitivity assay, with fluorescence measured using a Varioskan LUX multimode plate reader (Thermo Fisher Scientific). We recorded extract concentrations and processing information for each specimen. Progression to indexing and sequencing required an amplicon of the expected size after first-round PCR. Extracted DNA was stored at –20°C until PCR amplification.

The V3–V4 region of the bacterial 16S rRNA gene was amplified using the Bakt_341F (CCTACGGGNGGCWGCAG) and Bakt_785R (GACTACHVGGGTATCTAATCC) ~~primer pair~~⁴³ to target the V3–V4 region of the 16S rRNA gene. ~~Both primers carried Illumina overhang adapter sequences. We performed amplification using the Multiplex PCR kit (QIAGEN, catalog-primers~~⁴³, both carrying Illumina overhang adapters. First-round PCR was performed using TaKaRa Ex Taq DNA Polymerase (catalogue no. RR001B). Each reaction had a total volume of 50 µL and contained 5 µL of 10× buffer with Mg²⁺, 4 µL of dNTP mix (2.5 mM of each dNTP), 2 µL of each 10 µM primer stock, 0.25 µL of polymerase (5 U/µL), 5 µL of extracted DNA and 31.75 µL of nuclease-free water. The final primer concentration was 0.4 µM per primer and 0.2 mM per dNTP, with 1.25 U of polymerase per reaction. The PCR programme was: initial denaturation at 95°C for 3 min; 35 cycles of 95°C for 10 s, 65°C for 30 s and 72°C for 1 min; and a final extension at 75°C for 5 min.

First-round PCR products were assessed by electrophoresis on a 1% agarose gel. Samples showing an amplicon at the expected size of approximately 500 bp were classified as passing and were then indexed. Samples without a visible product were classified as failed; these were re-amplified with 8 μ L and then 12 μ L of DNA, with water reduced to 28.75 μ L and 24.75 μ L, respectively. DNA input was set by volume because most extracts had low or undetectable concentrations. Samples without a product after three attempts were classified as failed; the laboratory protocol also included re-extraction at the end of the workflow.

Dual indices and Illumina sequencing adapters were added in a second-round PCR using the QIAGEN Multiplex PCR Kit (catalogue no. 206143) ~~with an~~, following the Illumina 16S Metagenomic Sequencing Library Preparation protocol⁴⁴; initial activation at ~~95~~94°C for ~~15 min, 35 min~~; 8 cycles of 94°C for 30 s, ~~57°C for 90 s~~, 55°C for 30 s and ~~72°C for 90 s, and a~~ °C for 30 s; final extension at ~~72°C for 10 min~~, and ~~sequenced the resulting pools~~ °C for 5 min.

Indexed libraries were purified using Agencourt AMPure XP beads at a 0.8:1 bead-to-sample volume ratio, quantified using the Qubit dsDNA High Sensitivity assay, and assessed for fragment size using an Agilent 4200 TapeStation with the D5000 ScreenTape assay. Libraries were pooled by molarity calculated from concentration and fragment size in the Core Lab pooling calculation. Supplementary Section S2 specifies the pooling calculation and the remaining run-specific metadata gaps. Amplicon pools were sequenced on an Illumina NovaSeq 6000 ~~platform. We processed all samples using~~ an SP flow cell with a 500-cycle reagent kit to generate 2 $\times$ 250 bp paired-end reads. Sample processing and library preparation were performed in the PI's laboratory and sequencing was conducted at the Bioscience Core Lab at KAUST.

2.4 Sequencing data processing

We processed raw amplicon reads with nf-core/ampliseq (v2.14.0)^{45,46}, applying the DADA2 algorithm⁴⁷ for error correction and denoising, and assigning taxonomy against the SILVA database (v138.2)⁴⁸. The ~~dataset comprises submission records contain~~ 1,268 ~~sequencing libraries from run rows over~~ 1,238 ~~samples; sample identifiers, including~~ 30 ~~samples were sequenced twice, repeated identifiers~~. The feature-table export separately contains 29 identifiers with nonidentical paired columns; 28 retain both profiles after depth filtering. The data model distinguishes submitted runs, physical libraries and analytical profiles; the processing relationship of the paired export columns remains unresolved. This resulted in a canonical feature table of 351,472 amplicon sequence variants (ASVs) prior to contaminant removal, covering samples from Trips 1–5 and 24 control samples. ~~Of its 1,271 profiles, 1,247 are biological; quality filtering excludes the 24 control profiles and ten biological profiles with fewer than 1,000 reads, including one T1Dr1 run with 934 reads. This leaves 1,237 profiles, including 1,227 profiles from sites 1–60, for the companion ecology analysis. The complete table is distributed unfiltered. The sequencing inventory links all 1,271 canonical profiles to paired-read inputs: 1,013 from the July processing group and 258 from Trip 5. A further 33 July inputs fall outside the canonical feature table. The July execution used pipeline groups A and B. Processing specified the Bakt.341F/Bakt.785R primers, the NovaSeq option, DADA2 SILVA v138.2~~

assignment and skipped QIIME steps. The container digest, full denoising parameters, physical run allocation and profile-merging provenance remain incomplete for exact regeneration from raw reads.

~~We applied an assay-aware contamination audit following recommendations for low-biomass studies ⁴⁹. We pooled~~ We compared ASV prevalence in the 17 extraction day blanks that were processed on the same day as Trip 5 samples, and tested each ASV for enrichment in extraction blanks using a ~~eligible Trip 5 extraction blanks~~ with prevalence in the 217 batch-linked biological profiles ⁴⁹. Presence means a read count greater than zero. For each of the 351,472 ASVs, the one-sided Fisher's exact test (test compares present/absent profile counts in these two groups. ASVs absent from all eligible blanks receive a score of one. The exploratory candidate rule requires $p < 0.1$ (uncorrected), requiring presence in at least two blanks. ~~Because each extraction batch is represented by a single blank, within-batch prevalence could not be estimated. This resulted in a candidate set of~~, and higher prevalence in blanks than in biological profiles. It identifies 351 ASVs, applied candidates and applies their removal only to the 217 biological samples linked to those batches. We used seven positive control samples from Trips 1-3 to evaluate genus-level recovery, but not to train the contaminant classifier. The primary linked biological profiles. Prevalence is estimated across batches, each represented by one blank; the canonical table remains unfiltered.

Threshold sensitivity analysis also evaluated $p < 0.01$ and ~~the candidate set is used only in a bounded sensitivity analysis (Section 5)~~ $p < 0.05$, yielding 221 and 300 candidates with the two-blank rule. At $p < 0.1$, requiring one or three blanks gives 12,051 or 175 candidates. This sensitivity documents the role of the minimum-blank requirement. The control-recovery analysis separately uses four standards from Trips 2-3.

The Trip 4 screen applies the same rule independently to six mapped blanks and 95 biological profiles: one-sided Fisher $p < 0.1$, presence in at least two blanks and greater prevalence in blanks. It identifies seven candidates, whose removal is restricted to these 95 profiles. Together, the Trip 4 and Trip 5 screens cover 312 biological profiles.

~~We calculated genome-level functional potential inferred gene and pathway profiles from marker genes~~ with PICRUSt2 (v2.4.1) and used it for the companion analysis. ~~The execution manifest with the exact commands, ⁵⁰. Two executions used the Trips 1-4 amplicon output and Trip 5 output (330,830 ASVs). KO hidden-state prediction used the PIC method, edge exponent~~, seed and thresholds, is archived with the release0.5 and seed 100. EC/KO metagenome prediction required 100 reads; pathway inference used the EC predictions. The command sequence and output identities are included in the functional-prediction provenance described in Section 3. The Trips 1-4 input sequences/table and their checksums remain unavailable. The merged products contain 1,270 profile columns, whose mapping is distributed with the functional tables.

2.5 Taxonomy normalization and mapping

~~Taxonomic labels are text strings, and the same string~~ A taxonomic label can denote different taxa across databases and releases, and ~~one taxon can be referred to by~~

~~several strings a taxon can have several labels~~ ⁵¹. We therefore ~~map NCBI Taxonomy~~ ⁴¹ ~~identifiers to SILVA lineages alongside every assignment~~ retain the original SILVA lineage alongside a validated NCBI Taxonomy identifier or a lineage-scoped project identifier. Taxonomic assignments come from SILVA v138.2 (Section 2), and we treat the original classifier lineage and ASV identifier as the primary evidence. These assignments are seven rank fields (domain–species), and some are followed by a separate species assignment field and a numeric confidence. We assert the species from the ranked lineage and retain the additional species call in a provenance ~~ledger~~ ^{annotation}. We normalize the seven ordered rank fields by removing rank prefixes and representing empty and unclassified entries explicitly.

~~A fail-closed normalization step evaluates the historical~~ ^{The mapping procedure evaluates the} SILVA-to-NCBI candidates against a pinned NCBI Taxonomy release (the OBO Foundry NCBITaxon release 2025-03-13, built from the NCBI organismal taxonomy of 2025-03-01). An identifier is retained only when it resolves, has a compatible rank and label or documented synonym, and its ancestor chain is compatible with all retained higher-rank identifiers in that source lineage. A missing, ambiguous or incompatible candidate is replaced by a deterministic project class scoped to the complete truncated source lineage and rank, ~~rather than by the nearest ancestor's identifier, which would conflate a named child with its broader ancestor~~. Each decision records the source lineage and rank, candidate identifier, validation outcome, final identifier and reason. The output mapping covers every lineage represented in both the taxonomy export and the feature table; otherwise generation stops.

~~Existing project identifiers are retained only when all explicit historical uses share~~ ^{Project identifiers are scoped to} one normalized rank, label and ~~nearest explicit parent context; a reused identifier that acquires more than one corrected context is replaced in every affected use by contextual classes. Conflicting historical evidence is therefore contextualized rather than resolved according to file order~~ ^{parent context}. The mapping builder ~~emits corrected mappings, a decision ledger, a clean~~ ^{assigns contextual classes wherever those contexts differ. It produces the validated mappings, their decision provenance,} a project taxonomy module and ~~a manifest containing input and output checksums, and the abundance-ABox input/output checksums. The abundance generator accepts only a passing, checksum-matched manifest~~ ^{mappings that pass these checks.}

~~Queries retain the original SILVA lineage and independently validated NCBI or project identifiers. The mapping provenance reports rejected candidates and their reasons. Algorithmic checks establish consistency with the specified name, rank and ancestry rules; semantic mapping accuracy requires a separately annotated validation sample.~~

~~We evaluated cross-resource alignment to GTDB~~ ⁵² ~~and iNaturalist separately from this SILVA-to-NCBI layer, in an earlier exploration whose candidate tables are not part of the release. The conclusion that exploration supports is a negative one: homonyms, *Candidatus* names, placeholder taxa, GTDB split taxa, rank conflicts and database version changes can all produce identical labels for non-identical concepts, so label agreement alone does not establish extensional identity. The canonical release modules therefore contain no owl:equivalentClass assertions generated from those lexical candidates. Unreviewed GTDB and iNaturalist candidates are~~

~~excluded from the asserted crosswalk, and we claim no mapping precision or recall because a stratified manual gold standard has not been completed. This conservative representation permits queries over the original SILVA lineage and validated NCBI classes without turning lexical similarity, a shared label or a supplementary species call into unsupported taxonomic identity.~~

2.6 Shotgun-derived data

We distribute genome-relative-abundance profiles for 150 shotgun libraries and eggNOG-mapper annotations of the genome catalogue for comparison with amplicon-derived functional profiles in the ecology analysis. CoverM⁵³ reports each genome's abundance as a percentage of reads, retaining the unmapped fraction separately. The catalogue contains 991 genomes. We annotated the concatenated predicted proteome with eggNOG-mapper v2.1.12⁵⁴ on the KAUST Ibex cluster on 14 June 2026. The intersection of annotation and abundance data yields a 990-genome by KEGG Orthology copy-number matrix.

The ecology analysis compares whole-profile KO abundances in 125 libraries with matched amplicon profiles. Its marker-gene comparison uses 975 genomes and 119 matched libraries under the marker-specific annotation and abundance requirements. These data support downstream re-execution. Reproduction from raw shotgun reads additionally requires sequence accessions, genome and proteome catalogues, and the assembly, binning, dereplication and profiling workflows. These upstream inputs and parameters remain unavailable in the distributed resource.

2.7 PMA-treated aliquots

The PMA dataset contains nine treated/untreated pairs from three site-compartment groups at two Trip 5 campsites: C1 root-adjacent, C2 root-adjacent and C2 surface soil. For each group, three soil subsamples from one parent collection tube were prepared as separate slurries. Each slurry was divided into treated and untreated aliquots, giving three preparation pairs per parent tube. Suspensions used PBS at pH 7.4 and a 1:10 mass-to-volume ratio. Treated aliquots received 50 μ M PMAxx. Both treated and untreated aliquots incubated for 10 min in darkness and were illuminated at 450 nm for 35 min at 15–20 cm, with rotation at 15-minute intervals. Samples were frozen at -20°C ; DNA was extracted from 860 μ L using the PowerSoil Pro kit.

The paired marker-gene profiles describe treatment responses within these three parent specimens under matched dark and light exposure.

2.8 Environmental and climate data

Field environmental observations retain degrees Celsius, hectopascals (millibars) and percent relative humidity. Pressure magnitudes are multiplied by 100 and asserted in pascals (`obo:UO_0000110`) in the graph. Climate observations use degrees Celsius, millimetres and percent relative humidity, as applicable to each variable.

We retrieved historical climate data from the Open-Meteo archive endpoint³⁷ for each sampling site using its geographic coordinates, yielding 12,936 monthly ~~records encompassing~~ values: four variables over 49 months at 66 resolved sites, or 3,234

site-month rows. The variables are temperature, total precipitation, rainfall and relative humidity; ~~as well as annual records. Furthermore, field~~; annual summaries are distributed separately. We ran two separately configured acquisitions: the monthly product covers 1 January 2022 to 20 January 2026 and includes mean relative humidity, whereas the daily product covers 1 January 2022 to 1 February 2026 and reports temperature, total precipitation and rainfall. January 2026 is a partial month in the monthly acquisition. The acquisition metadata identify the query configuration and the retained climate values. Original response bodies, retrieval timestamps and the model/product version remain unavailable for the monthly acquisition. Reproduction uses the distributed climate observations; a new API request constitutes a new acquisition.

Daily rainfall exposures in the ecology analysis use the sampling-site coordinates and the distributed Open-Meteo time series. A five-product comparison combines Open-Meteo with CHIRPS, CMORPH, IMERG and NASA POWER; its provenance identifies the grid cells and 21 source datasets used. The daily exposures and monthly summaries are distinct data products with their own acquisition metadata.

Daily NASA POWER PRECTOTCORR precipitation covers sites 1–60 from 1 January 2022 to 31 December 2025 (87,660 site-days). Its original API request metadata were not preserved. Sites 51–55 share one POWER grid cell.

Field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO)⁴⁰.

2.9 XRF acquisition, processing and reconciliation

We acquired XRF data through two complementary workflows. During Trip 5, field researchers used ~~a Vanta handheld instrument to perform an Olympus Vanta VMR-XXX-G3-E handheld analyser (serial 841307) for~~ in-situ field measurements. The source log contains 106 entries across 59 sites; 71 complete sessions at 58 sites were retained, and nine of those sites have repeated complete sessions. The field ~~measurements do not refer to a specific specimen, so field and laboratory linkage is site-level rather than a same aliquot comparison~~ log identifies measurement sessions and sites. Field and laboratory observations are paired by site. Archived material from all five ~~expeditions campaigns~~ was subsequently measured in the laboratory. The laboratory inputs contain 547 selected sample records from Trips 1–4 and 178 from Trip 5 (725 records in total); ~~and all 725 are used in the analyses.~~

~~The provenance audit reproduced all 13,390 reported~~ Laboratory measurements used a Vanta handheld analyser in Fast Mode⁴². All 178 Trip 5 specimens used Fast Screening-He8mm with helium and an 8 mm diameter. Element and oxide-equivalent channels are retained separately. The analytical dataset uses the Fast Screening observations for V4Dr3 and V18Dr3. The exact gas and diameter settings for Trips 1–4 ~~cells under the current largest-positive-value rule and all~~ remain undocumented. Suppl. Section S3 describes acquisition and observation selection.

The laboratory acquisition metadata lack the exact model variant, serial number, measurement duration, tube settings, sample preparation, concentration units

and calibration results. Statistical-error percentages and detection-limit units are distinct from concentration units. We retain instrument-reported magnitudes with undocumented concentration units. At Site 4, ~~293 reported Trip~~ field readings targeted soil and a second substrate described as salt or oil; its substrate identity remains unresolved.

Repeated analyte entries are combined using the maximum positive value for Trips 1–4 and the last reported value for Trip 5. Repeated Trips 1–4 entries have identical values. Trip 5 ~~cells under the parser’s different last-reported-value rule. Neither rule is treated as instrument-prescribed, and the release discloses the two source-specific policies rather than implying a common calibrated procedure~~ contains repeated Al, Na and Si measurements for V14Dr1, for which the last reported value is retained. These rules define the 725-record laboratory dataset and are evaluated against the acquisition data in Technical Validation.

~~All XRF instrument channels (e.g. element, oxide) were mapped to a chemical identifier in a pinned ChEBI ³⁹ release and cross-checked against a PubChem ⁵⁵ snapshot. An executable audit rejects identifiers that cannot be found, that have been deprecated, or whose formula does not match the channel. The chemical audit evaluates 93 reporting channels against ChEBI release 247 (3 December 2025) and the PubChem property snapshot of 23 July 2026 ^{39,55}. It retains 79 supported ChEBI and 91 PubChem cross-references; unresolved channels have explicit null identifiers. Checks cover identifier existence, labels, formula, charge and element/oxide type. The P₂O₅ channel uses an oxide-equivalent reporting basis linked to P₄O₁₀ by empirical ratio. Light Elements remains an instrument-defined aggregate channel.~~

For a descriptive cross-workflow diagnostic, we took the median of repeated field sessions at each site and compared it with the Trip 5 Deep laboratory record at the 58 sites measured by both workflows. ~~Rank agreement was modest. We do not infer calibration between the two workflows, and no analysis treats field and laboratory values as being on the same scale. The diagnostic estimates site-level rank agreement. Calibration would require paired aliquots, documented units and reference materials. Technical Validation reports the observed correlations.~~

2.10 Archived-soil pH acquisition and admission

The archived-soil pH dataset contains admitted measurements linked to their physical specimens and assay sessions. The ecology analysis uses the same measurement values. We measured archived dry soil after sieving below 2 mm. Each assay used an independent 1 g scoop suspended in 2.5 mL 0.01 M CaCl₂. ~~Electrodes were calibrated~~ Suspensions were mixed on a HulaMixer for 30 min and stood for 30 min at room temperature. We used an accumet AB315 meter with a Fisher Scientific accumet electrode (catalogue no. 13-620-112), and recorded the reading when the meter displayed its green stability indicator. The electrode was calibrated each morning at two points (pH 7 and 10), and we recorded the electrode slope and pH-10 buffer read-back for every session. ~~Between samples, the electrode was rinsed along its full length with Milli-Q water, immersed in a separate container of Milli-Q water and dried by gently tapping with Kimwipes.~~ A measurement was admitted only if its session was

calibrated, the electrode slope was between 95% and 102%, and the read-back was within 0.1 pH units of nominal.

~~This resulted in~~ Admission additionally requires a numeric pH between 0 and 14 and an accepted measurement date on or before 3 August 2026. Assay dates must resolve unambiguously to calendar dates; observations with ambiguous or post-cutoff dates are excluded. Threshold endpoints are inclusive. Accepted measurements span 13 July to 2 August 2026. Of 1,168 ~~recordings of which specimen-assay entries~~, 712 measurements ~~were admitted~~. ~~The remainder met the admission criteria~~. The remainder consisted of 356 rows ~~that lacked~~ without a measurement, 45 marked as depleted, 36 with ambiguous dates, and 19 numeric rows with incomplete quality control records. ~~We neither inferred missing quality control fields nor imputed unavailable values.~~ ~~The workbook has no recorded session identifier~~. We therefore Rows with unavailable values or incomplete admission evidence retain their audit dispositions. We reconstructed 29 sessions from ~~the unique combinations of~~ trip, accepted ~~measurement~~ date, electrode slope, and pH-10 read-back, ~~as these groupings preserve the quality control context available in the source, but are inferred rather than recorded laboratory batch identifiers.~~

2.11 Schema validation

To ensure structural integrity within the RDF graph, we used Shape Expressions (ShEx) ⁵⁶. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived soil pH processes, values and sessions, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. This constraint application guarantees that all generated data conforms to the expected structural properties. We implemented this validation using the Apache Jena ShEx library (v4.10.0) invoked from custom Groovy scripts. The multi-gigabyte taxonomy ABox is excluded from this memory-bounded ShEx run; the exact generated file is instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process dataset-quality-value relationship.

2.11 Logical consistency (OWL reasoning)

We checked logical consistency using the ELK reasoner ⁵⁷. The ELK reasoner operates on the OWL 2 EL profile and detects unsatisfiable classes and contradictory axioms within ontologies. ELK implements a polynomial-time algorithm, facilitating the

consistency validation of large knowledge bases. However, this performance benefit necessitates a tradeoff in logical expressivity, as the OWL 2 EL profile does not support complex constraints such as universal quantification or inverse object properties found in ontologies like the Semanticscience Integrated Ontology (SIO). We employed ELK instead of more expressive reasoners (e.g., HermiT⁵⁸) because verifying the large knowledge graph we created with a more expressive reasoner proved impossible due to computational complexity. The current run is memory-bounded and excludes the multi-gigabyte taxonomy ABox and large reference modules, so a successful run establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. This approach enabled rapid continuous integration during ontology development.

2.11 Reproducible workflow

Release generation, validation and manuscript-derived statistics are orchestrated from one Nextflow entry point (`main.nf`), with separate local and cluster profiles that change execution infrastructure without changing process definitions. The workflow regenerates the site, environmental-measurement, sample, XRF, DNA, sequence-record, quality-control, laboratory-control and pH modules from their declared sources and requires every byte stream to match the staged release candidate, so the deposited modules and the manuscript statistics are produced from the same auditable run. All 156 admitted Trip 4 pH measurements used archived field replicate 2. Each admitted measurement is linked to its physical specimen and retains its original assay identifier. Matching with community profiles is at campaign, site and compartment level. The archived RDF protocol identifies an accumet AB315 with Probe 3 and automatic temperature compensation, and specifies mixing without fine grinding.

3 Knowledge representation

Table 4 Admitted archived-soil pH observations by campaign and compartment. Trip 4 measurements use field replicate 2.

Campaign	Surface	Subsurface	Root-adjacent	Total
1	52	56	49	157
2	5	6	12	23
3	54	50	52	156
4	50	52	54	156
5	30	57	133	220
Total	191	221	300	712

2.1 Knowledge representation

We developed a formal knowledge base to integrate the different types of data generated in this project. The semantic layer encodes the full provenance chain from physical sampling to digital data products, enabling cross-domain queries and ensuring data interoperability through shared vocabularies.

2.2 Architecture

The knowledge base is constructed in Description Logic⁵⁹ using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox defines the ontology of the knowledge graph using OWL 2 DL³⁵, while the ABox contains the instance data generated from the expedition. We deployed the complete knowledge base across three named RDF graphs: an ontology graph (17,025 triples) containing the TBox, a knowledge base graph containing the asserted instance data, and a taxonomy reference graph containing the aligned taxonomy module.

2.2 Prefixes used in listings

For brevity, the Turtle and SPARQL listings throughout the paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against the public endpoint. With these declarations in place, identifiers such as `rak:P100001` are abbreviated to `rak:P100001` and `sio:SIO-000291` to `sio:000291` without loss of information.

2.2 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO)³⁸ as represent the resource as a modular RDF knowledge base. The terminology module (TBox) defines classes and relations in OWL³⁵; the domain modules (ABoxes) describe sites, physical specimens, processes and data products. Shared identifiers connect these modules (Table 5). SIO supplies the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific domains, ensuring semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (see Section 7).

The canonical terminology module³⁸ The terminology declares 333 classes (297 local project classes and 36 referenced classes (32 SIO and four PATO⁶⁰). It also declares 20 object properties (11 project-local and nine SIO) and 35 datatype properties (34 project-local and one SIO). These are declaration counts from the staged canonical module, not counts of the imported ontologies or of whole-graph axioms. This compact semantic structure maintains precise alignment with our standardized schemas while minimizing redundant inferences. These counts apply to declarations in the terminology module. KG v3.0.1 supplies reviewed English definitions for all 342 project terms, recorded as SKOS and IAO definition annotations. Claude Sonnet (model identifier `claude-sonnet-5-5`) drafted the definitions from term labels, axioms

and dataset context; OpenAI Codex performed automated review and editing. The terminology identifies four reference ontologies as imports; observation modules are linked as dataset parts. Definition drafts, model provenance and review dispositions are distributed with the release.

2.1.1 Measurement ~~pattern~~ and provenance

We ~~model a quantitative measurement using SIO's ontology design pattern for measurement processes: a measurement is a process that generates~~ separate a measured quality from the information that quantifies it. A temperature quality belongs to a site; a measurement process produces a value for a specific quality of a target entity.

that quality. The value contains the numerical reading and, where documented, its unit. Counts, dimensional measurements and measurements with undocumented units use the appropriate datatype and unit constraints in their assigned shapes.

~~We identified all units with classes from the Units Ontology (UO) ⁶¹. Qualities are expressed using the Phenotype and Trait Ontology (PATO) ⁶⁰ and linked to SIO as subclasses of the Quality class. To illustrate this pattern, we provide an example represented in Turtle format. The example (Listing 1) demonstrates an environmental temperature measurement taken at a sampling site.~~

In words, an instance `rak:P000001` of a measurement process has as its target the sampling site `rak:1000001` and produces as output the quantity `rak:4000001`. That quantity carries the literal value 20.7, is reported in degrees Celsius (`uo:U0-0000027`), and refers to a temperature quality `rak:5000001` which is in turn illustrates this distinction for a field temperature. The process belongs to a site visit, the visit targets the site, and the measured quality is an attribute of the same site. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies that site. The process-to-visit path supplies the observation context. SIO's value-to-quality relation and its inverse support traversal in both directions.

2.1.2 Sampling pattern

~~A sampling event is a process that derives a sample from a feature of interest (the site).~~

~~Listing 2 illustrates the semantic representation of a surface soil sample collected during Trip 1.~~

~~Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output; the deep soil sample `rak:6000001` is in turn asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.~~

2.1.2 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO's general *Process-hasInput/hasOutput-Object* idiom and adds a specification link to a reusable protocol description.

Listing 1 Field temperature and its visit provenance.

```
@prefix rak: <https://rubalkhali.science/kb/RAK_> .
@prefix sio: <http://semanticscience.org/resource/SIO_> .
@prefix pato: <http://purl.obolibrary.org/obo/PATO_> .
@prefix uo: <http://purl.obolibrary.org/obo/UO_> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

rak:P000001 a rak:0000006 ;
    sio:000068 rak:3000001 ;
    sio:000229 rak:4000001 .
rak:3000001 a rak:0000003 ;
    sio:000291 rak:1000001 .
rak:4000001 a rak:0000010 ;
    sio:000221 uo:0000027 ;
    rak:2000003 "20.7"^^xsd:double ;
    sio:000215 rak:5000001 .
rak:5000001 a pato:0000146 ;
    sio:000011 rak:1000001 ;
    sio:000216 rak:4000001 .
```

Listing 3 demonstrates the formalization of a DNA extraction process. Sampling processes output material-sample collections. Collection membership links each physical replicate to the sampling event; the specimen's derivation edge also links it directly to its site.

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:4900001`) which is itself quantified through a separate measurement process following the pattern of Equation 1. A protocol is a reusable information specification, and a protocol execution is a process.

2.1.2 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the merged taxonomy (see Section 2) to assign specific NCBI or GTDB classes as the target of each observation. Listing 4 shows how we map an entire sample's feature table row into the knowledge graph. Extraction consumes soil and produces DNA; library preparation consumes DNA and produces an amplicon library; sequencing consumes the library and produces a digital FASTQ dataset. Quality-control processes consume FASTQ datasets and produce direction-specific metrics. Suppl. Sections S1.2–S1.7 give the corresponding graph examples.

For instance, the sequence analysis for a specific sample yields an absolute sequence count associated with a taxonomic class from the NCBI Taxonomy. We construct an abundance quality that targets the explicit taxonomic identifier, binding the observational count back to the biological workflow. Formally, the pattern instantiates SIO's *Collection-hasMember-Thing* and *Quantity-refersTo-Quality* idioms:

Site annotations use existential restrictions to relate a site to an ENVO biome or environmental feature. Such a restriction specifies a filler of the named class. The source annotation metadata provides the recorded assignment and its provenance.

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes a FASTQ dataset(`rak:7900001`) and produces the taxonomic dataset `rak:7290001`; that dataset has as one of its members the quantity `rak:4400001`, which records an absolute read count of 15 and refers to an absolute abundance quality that is in turn an attribute of *Pseudomonas* (`obo:NCBITaxon:286`). One row of a feature table is therefore expanded into a four-node sub-graph that preserves both the classifier output and its provenance. Laboratory controls are occurrence-specific materials with processing roles. A control-processing execution consumes the control, realizes its role and belongs to a documented processing group. Trips 4–5 use trip-specific extraction-day batches. Trips 1–3 use per-kit controls, and Trips 1–2 include shared extraction days. Occurrence-specific identifiers and the identity metadata distinguish reused control labels. Confirmed, provisional, unresolved and inapplicable dispositions retain their supporting evidence. The D6300 whole-cell and D6322 purified-DNA standards enter the process chain at extraction and PCR, respectively.

2.1.2 XRF geochemistry pattern

2.1.2 Taxonomic and chemical observations

For each profile, the taxonomy generator aggregates ASV counts at seven ranks from domain to species, retaining the rank-truncated classifier lineage and its validated NCBI or project identifier. Each workflow produces fourteen datasets: read counts and relative read abundance for each rank. The relative value divides the aggregated count by the profile's total reads; explicit unclassified lineages retain their contribution to that denominator. A taxon-rank entry adds a value and a quality linked to shared dataset, workflow, taxon and FASTQ objects. These links preserve the profile context during retrieval. The taxon IRI is used as an individual in the abundance graph and as a class in the mapping module. OWL 2 punning gives these uses separate semantics: observation links identify the measured taxon, while class-level axioms supply the reference ancestry. The generator performs the rank aggregation explicitly.

We model XRF analysis as XRF processes have several concentration-value outputs. A laboratory process takes an archived specimen as input; a field process targets a process that targets a soil sample and produces multiple measurement values, each corresponding to the concentration of a specific chemical analyte. We map each analyte to the Chemical Entities of Biological Interest (ChEBI) ontology³⁹ as well as to the PubChem database⁵⁵ (site and retains the instrument test identifier. The chemical mapping supplies supported ChEBI and PubChem cross-references for reporting channels, including oxide-equivalent channels. Unit documentation and

acquisition scope are described in Section 2.9). Out of 93 monitored geochemical parameters, we mapped 80 analytes directly to corresponding ChEBI classes and aligned 92 parameters to PubChem Compound Identifiers. The XRF pattern is a specialization of the measurement pattern (Equation 1), with the target restricted to a soil sample and the produced quantity bound to an analyte-specific quality:

Listing 5 illustrates an XRF measurement for a specific analyte.

In words, the XRF measurement process `rak:P300001` targets the soil sample `rak:6000001` and produces the quantity `rak:4300001`, which records a concentration of 58.2 percent (`uo:U0-0000187`) and refers to an analyte-specific quality `rak:5300001`. Because the analyte quality is typed with a dedicated subclass per element, the same pattern instance can be reused unchanged across all 91 geochemical analytes by varying only the quality class IRI. Archived-soil pH processes consume and target the assayed specimen, produce a pH value linked to its acidity quality and belong to a reconstructed measurement session. Measurements from different field replicates retain separate physical identities and are matched at campaign, site and compartment level (Section 2.10).

2.1.3 Sampling site and biome assignment pattern

2.1.3 Formal patterns

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. We use both an existential and a universal restriction so that the asserted biome both holds and is the only admissible value, which preserves the closed-world intent of the field record. The following Description Logic patterns⁵⁹ specify the entity relationships used by the resource. The notation uses $\sqsubseteq$ for subclass inclusion, $\sqcap$ for conjunction, $\exists R.C$ for at least one relation R to a member of class C , and $\forall R.C$ to restrict all R fillers to C . Structural validation checks the corresponding data constraints. Suppl. Sections S1.2–S1.7 provide concrete RDF examples.

The sampling pattern links a process to its target site and output collection, whose members are physical samples.

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (2)$$

$$\text{SampleCollection} \sqsubseteq \text{Collection} \sqcap \forall \text{hasMember}.\text{Sample} \quad (3)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget}.\text{Site} \sqcap \exists \text{hasOutput}.\text{SampleCollection} \quad (4)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom}.\text{Site} \quad (5)$$

In words, the sampling site `rak:1000001` carries a label, a WKT geometry, and is asserted both to belong to the ENVO desert biome and to exhibit a sand-dune environmental feature. The combined existential and universal restriction on `hasBiome` (Equation 24) ensures that any reasoner treats the asserted biome as the only admissible value for that site. A protocol execution links an information specification to the entities consumed and produced by the process.

2.1.4 Sequencing workflow pattern

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (6)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (7)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy.Protocol} \quad (8)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput.Entity} \sqcap \exists \text{hasOutput.Entity} \quad (9)$$

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process-hasInput/hasOutput-Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of Equation 6. The abundance pattern connects a sequence-analysis workflow to a collection of values and to the taxonomic qualities those values quantify.

~~Read as a chain, the DNA extraction process `rak:P200001` consumes the soil sample `rak:6000001` and produces the DNA extract `rak:6800001`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6900001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7900001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the full provenance from any node.~~

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (10)$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (11)$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (12)$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (13)$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \quad (14)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (15)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \quad (16)$$

2.1.4 Sequencing quality control pattern

XRF specializes the measurement pattern through a specimen input for laboratory measurements and a site target for field measurements.

~~Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of Equation 1. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.~~

$$\text{XRFProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \quad (17)$$

$$\text{LabXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasInput.SoilSample} \quad (18)$$

$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \quad (19)$$

$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AnalyteQuality} \quad (20)$$

The sequencing pattern composes extraction, library preparation and sequencing through their shared inputs and outputs.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (21)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{DNAExtract} \quad (22)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (23)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput}.\text{DNAExtract} \sqcap \exists \text{hasOutput}.\text{AmpliconLibrary} \quad (24)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (25)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput}.\text{AmpliconLibrary} \sqcap \exists \text{hasOutput}.\text{FASTQDataset} \quad (26)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \quad (27)$$

$$\text{FASTQDataset} \sqsubseteq \forall \text{hasMember}.\text{SequenceRead} \quad (28)$$

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7900001` as input and produces the measurement value `rak:4500001`, which records a GC content of 54.2 percent and refers to a forward-read GC-content quality `rak:5500001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation. Quality-control processes attach measurement values to qualities of the source sequence dataset.

2.1.4 **Laboratory-control pattern**

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (29)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput}.\text{FASTQDataset} \sqcap \exists \text{hasOutput}.\text{QCMeasurementValue} \quad (30)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf}.\text{QCQuality} \quad (31)$$

$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf}.\text{FASTQDataset} \quad (32)$$

A laboratory control is an occurrence-specific material entity, not a property of an expedition and not the string used to label a library. Each physical control carries its own instance of a positive, negative, extraction-blank or PCR-blank role, and a laboratory process realizes that role while consuming the material as input and target and producing a derived material or dataset. Processes are grouped into batches, and, because an extraction day can contain samples from several expeditions, neither the control nor its batch is linked directly to a trip.

2.1.4 **OWL and RDF implementation**

Groovy scripts using OWL API v5.1.20⁶² generate the modules. The modular organization separates domain data from shared terminology⁶³. OpenLink Virtuoso⁶⁴ provides RDF storage and SPARQL query execution.

Listing 9 instantiates the pattern for an extraction blank. The blank material `rak:C000001` bears an extraction-blank role `rak:R000001` through `sio:000228` (*has role*); the extraction process `rak:P400001` takes the blank as input and target,

~~realizes the role through `sio:000355`, KG v3.0.1 serves the asserted union. A separate reasoning workflow uses the bundled rule program, v1.5.0, to compute consequences of a specified set of positive RDF rules. These rules expand equivalent classes and properties, subclass and subproperty hierarchies, instance types, domains, resource-valued ranges, inverse and symmetric properties, and is part of the extraction batch `rak:B000001`. Asserting the role as an individual rather than as a class membership keeps each physical control distinct even when labels are reused across expeditions.~~

transitive properties. The schema audit generates rules for each nonempty finite property-chain pattern in the loaded schema, including inverse operands. Additional rules introduce and classify value restrictions, classify existential restrictions from existing edges and typed fillers, propagate universal restrictions to existing resource fillers, and handle intersections, union membership and resource enumerations. All conclusions use existing RDF terms. The schema audit lists the loaded imports and the constructs outside this rule set.

~~Uncertain identity, batch membership and assay applicability are represented as evidence-bearing statements with an explicit confirmed, provisional, unresolved or inapplicable status rather than as ground truth, so a reused or ambiguous label never merges two physical controls. Whole-cell D6300 and purified-DNA D6322 controls enter the workflow at different processes, which lets the graph distinguish extraction and lysis evaluation from PCR, library and sequencing evaluation. The control module contains 489 entities and 3, 800 triples, including 51 material-specific role individuals and 47 sequence occurrences. The asserted graph contains the deduplicated union of the loaded modules. Rules read this graph together with a separate inferred graph and insert only conclusions absent from both. Each update selects a bounded batch of distinct conclusions; each rule repeats until its next batch adds zero triples. The complete schedule repeats until every rule adds zero triples in one full pass. This stopping criterion defines the least fixed point of the emitted rule program.~~

The controller preserves the rule text, partition inventories, derived-triple counts, runtime versions and input identifiers. Query truncation, SQL errors and exhaustion of the pass limit terminate the run with a failure status. After convergence, separate diagnostics test explicit disjoint-class and complement clashes, membership in `owl:Nothing`, and incompatible laboratory-control roles. These diagnostics and the finite rule set define the scope of operational entailment validation.

2.1.5 Archived-soil pH pattern

2.1.5 Structural constraints

~~Archived-soil pH instantiates the generic measurement pattern of Equation 1 but binds it to an archived specimen rather than to a field site. The measuring process consumes and targets an existing specimen, outputs a single pH value, is specified by the `CaCl2` protocol, and is part of a dated measurement session. The value is the measurement value of an acidity quality borne by the same specimen. ShEx shapes⁵⁶ check selected nodes for required types, properties, cardinalities and datatypes. The sampling-site shape requires a site type, label and geometry predicate. Its unconstrained geometry~~

object checks presence; validation of the WKT datatype and coordinate syntax is a separate requirement. Additional type assertions are accepted independently of how they were obtained.

Listing 10 shows a single admitted pH observation. The specimen `rak:6000001` bears the acidity quality `rak:5600001`; the value `rak:4600001` carries the literal 7.8, has unit `UO:0000196`, and points back to the quality with `sio:000215` and to its producing process with `sio:000232`. The process is specified by the `CaCl2 protocol`. The sample shape requires a replicate type, a label, a source identifier and an IRI-valued derivation edge. The DNA concentration shape checks its numeric value, nanograms-per-microlitre unit and is part of the session `rak:B100001`. Rows that fail source, date, specimen or quality-control admission remain in the audit table and are not asserted as measurements (Section 2.10) quality/process links. The XRF-process shape requires a protocol, one or more values and an input or target IRI. Checking the class of that input or target requires a corresponding target shape or scientific-invariant check. These open shapes admit predicates outside their declared constraints; the negative fixtures test the specific failures reported in Technical Validation.

2.2 OWL and RDF implementation **Schema validation**

We generated the OWL/RDF implementation using custom Groovy scripts utilizing the OWL API 5⁶². The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata checked RDF structure using Shape Expressions (ShEx)⁵⁶. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control metrics, laboratory controls, archived-soil pH, and taxonomic abundances into individual ABox modules⁶⁵.

We use OpenLink Virtuoso⁶⁴ for RDF storage and query answering. In a development benchmark, a project-specific forward-rule program applied selected RDFS/OWL rules for subclass, subproperty, domain, range, inverse, symmetric, equivalence and transitive property assertions. The program used three scheduled passes, including only a one-hop transitive pass. It did not demonstrate convergence to a fixed point and did not implement arbitrary OWL property chains. We therefore describe this output as a *selected rule expansion*, not as complete OWL-Horst⁶⁶ closure, and we report exact module-level entity counts (Section 3) rather than historical asserted or inferred whole-graph triple counts.

2.3 Schema definition

We used Shape Expressions (ShEx)⁵⁶ to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph's structure, ensuring data quality and compliance before loading it into the database.

Our ShEx implementation defines shapes for Sampling Sites, Soil Samples, Environmental Measurements, DNA Extracts, XRF processes, Sequencing Datasets, laboratory controls, and records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived-soil pH observations. These shapes guarantee that essential attributes are invariably present. For instance, Listing 11 demonstrates how the schema enforces that every Sampling Site is explicitly typed and labeled, while capturing geographic coordinates and an optional textual descriptionpH processes, values and sessions, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. Each passing test establishes conformance to the assigned shape's specified constraints. We implemented this validation using the Apache Jena ShEx library (v4.10.0) invoked from custom Groovy scripts. The multi-gigabyte taxonomy ABox is excluded from this memory-bounded ShEx run; the exact generated file is instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process-dataset-quality-value relationship.

2.3 Logical consistency (OWL reasoning)

In words, We ran ELK v0.4.3 ⁵⁷ through OWL API v4.5.26, loading the tractable modules into a merged ontology. This operational check processes the constructs supported by ELK. The modules also contain universal restrictions, inverse properties and other constructs outside that support. An explicit OWL profile report and checked projection are required to establish the logical scope of the result. The run excludes the multi-gigabyte taxonomy ABox and large reference modules. We report the checked subset and the shape <#SamplingSiteShape> requires that every conforming node carry an explicit type assertion to `rak:0000002` (Sampling Site) and an `rdfs:label`; it permits any number of additional inferred types and optionally allows a free-text description. Coordinates are mandatory in the shape, and all 70 sampling-site individuals in the released module satisfy it. Records that omit the mandatory type, label or geometry are rejected before they reach the triple store, and a deliberately coordinate-free fixture in the negative-fixture suite (Section 4) proves that the rejection fires.

The remaining shapes reuse the same closed-world idiom but encode the domain-specific provenance and measurement contracts that a type assertion alone cannot express. Listing 12 shows the soil-sample shape: a conforming sample must be explicitly typed as a replicate, carry a label and a stable source identifier, and be derived from a sampling site. The derivation edge is what lets a query traverse from a specimen to its site without joining through the sampling eventreturn value separately from structural validation in Technical Validation.

2.4 Reproducible workflow

The DNA-extract and XRF shapes enforce the canonical measurement reification (split quality and value) of Equation 1. Listing 13 requires the concentration value to carry a numeric concentration, the nanograms-per-microlitre unit, and links back to both its quality and its producing measurement process, so a value that is detached from its process or that uses the wrong unit is rejected. Nextflow workflow orchestrates module generation, validation and manuscript-derived statistics. Local and cluster profiles configure execution infrastructure. The versioned KG release workflow regenerates the scientific modules, checks their hashes against the frozen manifest, loads a fresh asserted database and validates source-derived query results before switching the public service. Upstream sequence processing has its own sample inventories, workflow parameters and environments.

The XRF shape encodes the two acquisition workflows directly: a conforming XRF process must either take a collected specimen as input (the laboratory workflow) or target a recorded sampling site (the field workflow), and it must produce at least one value. This disjunction is what allows one shape to cover both workflows while still rejecting a process that has neither an input nor a target, the defect exercised by one of the negative fixtures.

The release manifests identify fixed source artifacts and generated modules. Reproduction has three stages: downstream analyses from curated inputs, generation of those inputs from primary records, and replication of the field/laboratory measurements. The package’s workflow evidence covers the executed stages; missing upstream sequence-processing inputs and acquisition metadata are stated in their respective Methods subsections. Versioned inputs and execution records identify the data and software used for the reported checks.

3 Data records

The resource is distributed as a single directory tree in which every payload file is declared in a machine-readable manifest (`PRE-RELEASE-MANIFEST.tsv`) with its byte size, SHA-256 checksum, category, release status and licence gate. The raw sequencing data are deposited in the European Nucleotide Archive (ENA) under the umbrella project accession PRJEB104209, with amplicon sequencing under project PRJEB106069. The knowledge graph is accessible via a public SPARQL endpoint at [https://www.ebi.ac.uk/ena/ena-sparql](#) and through a web interface at [https://www.ebi.ac.uk/ena/ena-sparql](#), and all source code for data processing, ontology generation, and validation is maintained at [https://github.com/ebi-europe/ena-sparql](#).

3.1 Knowledge-graph modules

The primary artifacts are the knowledge-graph modules, distributed as OWLKG v3.0.1 contains 45,707,660 asserted triples from 16 manifested modules and is archived at Zenodo ⁶⁷. The deposit provides the RDF/XML and Turtle files and structured by data type (modules in Table 5). The base ontology (`rubalkhali.owl`) holds the terminology; each remaining module holds the instance data for one domain and shares the same; a compressed N-Quads export of their deduplicated union, and code to load

the modules and reproduce that export. Shared site, specimen and process identifiers ~~so the modules can be loaded together into any RDF store or used individually connect~~ observations across modules. The release manifest identifies each deposited file and its checksum.

3.1 ~~Tabular data products~~

Table 5 The 16 RDF modules included in KG v3.0.1.

Module	Content
rubalkhali.owl	Base ontology (TBox)
rubalkhali_sites.owl	Sampling sites with coordinates and biome annotations
rubalkhali_samples.owl	Soil and plant specimen individuals with compartment types
rubalkhali_measurements.owl	Field environmental measurements and climate records
rubalkhali_xrf.owl	XRF measurement values from the field and laboratory workflows
rubalkhali_dna.owl	DNA extracts with concentration data
rubalkhali_sra.owl	Sequencing run records linked to ENA accessions
rubalkhali_qc.owl	Sequencing quality metrics (GC content, duplicate rate, sequence counts)
rubalkhali_controls.ttl	Laboratory-control materials, roles, batches and sequence occurrences
rubalkhali_ph_eq_ph_shared_v1_0_1.ttl	Archived-soil pH measurements, sessions and reconciled specimen links
rubalkhali_taxonomy_abox.ttl	Taxon observations over feature profiles
ecosystem_module.ttl	Vocabulary supporting canonical taxonomic mappings
sio.owl	Semanticscience Integrated Ontology v1.59
envo.owl	Environment Ontology, 20 October 2025
pato.owl	Phenotype and Trait Ontology, 14 May 2025
uo.owl	Units Ontology, 25 May 2023

~~Alongside the graph, the deposit distributes the curated tabular data products from which the modules are generated: the versioned source ledger linking site and trip to specimen, extraction, library, sequence run, quality-control decision and derived community; the canonical Trips 1-5 amplicon feature table and its taxonomy mapping; the correction and alias ledgers that record every deterministic metadata repair without overwriting source values; the curated field environmental table; the field and laboratory XRF tables together with the audited chemical-mapping ledger; and the versioned~~ The data repository provides the specimen catalogue, assay tables and analysis inputs. The catalogue links specimens to collection visits, extractions, libraries and sequence runs. Tabular products include amplicon counts and taxonomic

mappings, field environmental observations, laboratory and field XRF values, and archived-soil pH ~~dataset measurements~~ (EQ-PH-SHARED-v1.0.0. ~~The four derived companion analysis inputs (.1).~~ The repository also provides the paired PMA table, the CoverM genome-abundance profiles, the eggNOG annotations and the measured-function input archive) are distributed as records for downstream statistical re-execution rather than as graph modules.

3.1 ~~Validation schemas, queries and evidence~~

The deposit includes the artifacts needed to re-validate the resource: the ShEx shapes that define the structural contract for each entity family, the positive and negative structural fixtures, the archived competency SPARQL queries, and ~~profiles, 150 CoverM profiles, eggNOG annotations for the machine-readable evidence bundles (reconciliation summaries, audit tables, validation records and checksum manifests) produced by the release workflow. A companion data dictionary (metadata/DATA-Dictionary.tsv) records the field names~~ 990-genome analytical subset, and six source tables for the MAG-derived versus PICRUSt2 comparison. Tables are supplied in delimited text formats; larger collections are compressed. The climate inventory identifies daily inputs; monthly climate summaries retain separate acquisition metadata.

The data dictionary defines fields, data types, units, ~~and~~ missing-value conventions and definitions for the curated ~~canonical tabular records~~. Quantities carry units only where a unit is documented at the source: environmental source tables retain degrees Celsius, hectopascals and percent relative humidity; in the graph, pressure magnitudes are asserted in pascals, climate records use degrees Celsius and millimetres, and DNA concentrations are in nanograms per microlitre. XRF concentrations carry no documented unit assertion and must not be treated as percentages.

The data and source code to generate the Rub' al-Khali knowledge graph is ~~available at~~ tables. Site and campaign identify collection visits; specimen and assay identifiers distinguish physical material from its measurements. Control roles, processing batches and quality-control dispositions support selection of observations for reuse. Missing values and unresolved mappings retain the conventions documented for each table. The file inventory gives locations, sizes, checksums and licensing status. A separate bulk-data inventory gives retrieval information for larger downloads. The repository also supplies validation schemas, SPARQL queries, expected query outputs and the source comparisons described in Technical Validation.

4 Technical validation

We validated the knowledge graph using four complementary approaches, each targeting a distinct aspect of data quality: competency question evaluation via SPARQL queries, structural validation using Shape Expressions (ShEx), logical consistency checking with the ELK reasoner, and usability assessment through a web portal. Technical validation separates source reconciliation and measurement checks from structural constraints, executable queries and the operational ELK check. The evidence bundles identify the input snapshot and checks executed.

4.1 Competency questions and SPARQL queries Assay validation

To validate whether the full knowledge graph can be used to answer queries about the Rub' al-Khali, we defined competency questions expressed as SPARQL queries⁶⁸ with known expected answers. We automated this validation process to enable regression testing during ontology updates. The validated competency questions include retrieving The PowerSoil Pro extraction-control profile contained 1,165 ASVs, with 84% of reads assigned to ASVs shared with biological profiles. Its nearest canonical profile was 46Dr1, with Bray-Curtis dissimilarity 0.39. We excluded this control from the contaminant screen. A library-order test compared nearby library numbers with different-site pairs matched by geographic-distance decile and compartment, using 999 library-order permutations. The Trip 1 and Trip 3 Bray-Curtis contrasts were -0.003 ($p = 0.15$) and $+0.006$ ($p = 0.99$), respectively. These contrasts summarize the tested preparation-order association.

To assess agreement between field and laboratory XRF measurements, listing sampling sites with coordinates, extracting DNA extraction data, tracing provenance from soil samples to sequencing runs, and querying taxonomic composition. The competency questions are designed to test questions that are needed for operating the website, and to support different types of queries within our dataset. Repeated field sessions were reduced to site medians and paired with the Trip 5 subsurface laboratory observation at 58 sites. Among 12 analytes with positive values in both workflows at more than 10 paired sites, the median Spearman correlation was 0.21 (range -0.40 to 0.48). This diagnostic describes site-level rank agreement among the retained positive pairs.

4.2 Competency questions and SPARQL queries

Competency questions define the intended retrieval tasks (Table 6 lists the competency questions we predefined for validation. Each is expressed as a SPARQL query with a known expected answer and is run as a regression test whenever the knowledge graph is regenerated. Together they cover every core module: geochemistry (CQ1, CQ4), the site catalogue and its biome grounding (CQ2, CQ3), . The archived query workload covers site, geochemical, molecular-provenance and taxonomic data. The Site 10 XRF query has an archived 46-row result. Source-derived expected result tuples also specify the complete 48-observation genus profile for run ERR16061083, 360-row environment-taxon join, 70 site coordinates and six PCR no-template control roles. The release validator compares full result multisets, preserving duplicate rows and requiring exact identifiers. Numeric comparisons use absolute and relative tolerances of 10^{-12} ; coordinate comparisons use an absolute tolerance of 10^{-10} degrees. The query comparisons test the molecular provenance chain (CQ5, CQ6), links between observations and their specimen, process and site context.

For KG v3.0.1, release validation verifies all 16 input-module hashes and parses the complete 45,707,660-quad export, checking every graph context. The terminology validation checks the term axioms and one definition per annotation property for each of 342 project terms. Public-release checks verify equivalent ontology content in

[RDF/XML, Turtle, JSON-LD and ~~taxonomic composition \(CQ7\)~~N-Triples](#), resolution of the four reference imports, manifest integrity, the live asserted-graph count and the deposited artifacts. Section 3 identifies the validation evidence and its input versions. Structural tests evaluate the scientific modules separately.

Table 6 Competency questions defining the retrieval workload.

ID	Competency question
CQ1	What are the XRF analyte concentrations measured at a given sampling site?
CQ2	Which sampling sites are recorded, and what are their GPS coordinates?
CQ3	Which biome and environmental features are assigned to each sampling site?
CQ4	What are the instrument-reported Light Elements values across the XRF measurements?
CQ5	Which DNA extraction records (extract, concentration, protocol) exist for a given sample?
CQ6	What is the full provenance chain from a soil sample to its sequencing run?
CQ7	What are the complete genus-level read counts and relative read abundance for run ERR16061083?

~~We show several example SPARQL queries.~~ Listing 2 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 2 SPARQL query to retrieve field XRF measurements for Site 10. The query starts from the field-XRF measuring processes whose target is the site and walks to the individual analyte concentrations.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?processLabel ?siteLabel ?analyte ?concentration
WHERE {
  ?process a rak:0000025 ;
    rdfs:label ?processLabel ;
    sio:000291 ?site ;
    sio:000229 ?value .
  FILTER(CONTAINS(LCASE(STR(?processLabel)), "field xrf"))
  ?site rdfs:label ?siteLabel .
  FILTER(?siteLabel = "Site 10")
  ?value sio:000215 ?quality ;
    rak:2000012 ?concentration .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000032)
}
ORDER BY ?processLabel ?analyte
```

~~Read in prose, the query first selects every soil sample whose label matches Site 10, then walks back from each measurement value to its producing process and forwards from the sample through its quality and value to the recorded concentration and unit. The final filter excludes the Light Elements pseudo-analyte (rak:0000029)~~

~~so that only individually quantified elements appear in the result, and the query~~ The query selects field-XRF processes targeting Site 10, then retrieves each output and its analyte quality. It returns one row per process~~and analyte combination~~/analyte combination, excluding the aggregate Light Elements channel. The concentration units are undocumented.

The query returned 46 analyte-value rows for Site 10: 24 from Test 5847 and 22 from Test 5848. Validation compares the returned identifiers and values with the expected observations from the site and XRF data.

Listing 3 retrieves ~~taxonomic abundance profiles~~the complete genus profile for run [ERR16061083](#), joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 3 Complete genus-level profile for run ERR16061083, ordered by descending read count and lineage.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?proc ?fastq ?runLabel ?lineage ?taxon ?taxonLabel
       ?count ?relativeAbundance
WHERE {
  VALUES ?runLabel { "FASTQ dataset for ERR16061083" }
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs, ?dsRel .
  ?fastq rdfs:label ?runLabel .
  ?dsAbs a rak:0000074 ; rdfs:label ?absLabel ;
        sio:000059 ?valAbs .
  ?dsRel a rak:0000075 ; rdfs:label ?relLabel ;
        sio:000059 ?valRel .
  FILTER(STRENDS(STR(?absLabel), "(Genus)"))
  FILTER(STRENDS(STR(?relLabel), "(Genus)"))
  ?valAbs rak:2000026 ?count ;
        rak:2000025 ?lineage ;
        sio:000215 ?qualAbs .
  ?valRel rak:2000020 ?relativeAbundance ;
        rak:2000025 ?lineage ;
        sio:000215 ?qualRel .
  ?qualAbs sio:000011 ?taxon, ?fastq .
  ?qualRel sio:000011 ?taxon, ?fastq .
  ?taxon rdfs:label ?taxonLabel .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?runLabel DESC(xsd:double(?count)) ?lineage ?proc
```

~~In words, the query identifies every bioinformatic workflow that produces both an absolute and a relative taxonomic dataset for the same FASTQ input, then for each absolute observation it retrieves the read count, the associated taxon label, and the matching relative abundance from the parallel relative dataset. Results are grouped by sequencing run and ordered by descending count, so the most abundant taxa per run appear first. The query binds both count and relative-abundance values to the same workflow, source FASTQ dataset, taxon and classifier lineage. Dataset labels encode rank through the suffix (Genus). The query uses that emitted convention explicitly and returns all 48 observations for the named run. The source-derived~~

expected table contains the complete profile, including each classifier lineage. The all-run query retrieves profiles from a locally loaded snapshot. That unrestricted export is a long-running operation. The environmental example below additionally binds field temperature and specimen collection to the same site visit.

~~We added an additional query (Listing 4) to retrieve specific taxa abundances alongside environmental constraints, demonstrating the ability to cross-reference biological and environmental properties.~~

Listing 4 SPARQL query to retrieve *Pseudomonas* relative abundance at sites with high temperature.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?visit ?visitLabel ?site ?siteLabel ?sample ?fastq
       ?temperatureProcess ?tempVal ?temp ?lineage ?relAbundance
WHERE {
  ?visit a rak:0000003 ; rdfs:label ?visitLabel ;
        sio:000291 ?site .
  ?site rdfs:label ?siteLabel .
  ?temperatureProcess sio:000068 ?visit ;
        sio:000229 ?tempVal .
  ?tempVal rak:2000003 ?temp ; sio:000215 ?tempQual .
  ?tempQual a pato:0000146 ; sio:000011 ?site .
  FILTER(?temp > 35.0)
  ?sampling a rak:0000024 ; sio:000068 ?visit ;
        sio:000291 ?site ; sio:000229 ?collection .
  ?collection sio:000059 ?sample .
  ?ext sio:000230 ?sample ; sio:000229 ?dna .
  ?libPrep a rak:0000065 ; sio:000230 ?dna ;
        sio:000229 ?lib .
  ?seq a rak:0000066 ; sio:000230 ?lib ;
        sio:000229 ?fastq .
  ?bioinf a rak:0000071 ; sio:000230 ?fastq ;
        sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ; rdfs:label ?datasetLabel ;
        sio:000059 ?valRel .
  FILTER(STRENDS(STR(?datasetLabel), "(Genus)"))
  ?valRel rak:2000020 ?relAbundance ;
        rak:2000025 ?lineage ; sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286, ?fastq .
}
ORDER BY ?visit ?sample ?fastq ?tempVal ?lineage
```

Each result retains the collection visit, specimen, sequence dataset, temperature observation and genus lineage. A site's visits are joined through their process provenance. This provides an environmental temperature associated with the collection visit of each returned microbial profile.

~~In words, the query first selects every sampling site at which the recorded environmental temperature exceeds 35 °C, then for each such site it walks the full provenance chain — soil sample, DNA extract, library preparation, sequencing run, FASTQ dataset, bioinformatic workflow, relative abundance dataset — down to the relative abundance value of *Pseudomonas* (ncbitaxon:286). Each row of the result therefore reports a single hot site together with its corresponding *Pseudomonas* relative~~

abundance, demonstrating a join that crosses the environmental, geochemical, and biological domains in a single SPARQL query. Regression fixtures execute both queries on deliberately confusable inputs. The taxonomy fixture assigns the same genus different abundances in two profiles and adds a species-rank observation; the query returns exactly the two matching genus observations. The visit fixture gives one site two visits at 40 and 20°C; the temperature-filtered query returns only the specimen collected during the 40°C visit. These tests check the implemented joins; release-level validation additionally requires expected result tables for the complete modules.

4.3 Source reconciliation and measurement validation

The ~~executed~~ archived evidence workflow produced a ~~row-level sample ledger~~ specimen catalogue and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. A ~~site-alias gate~~ site-identity check independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement, accounting for all 36 ~~affected source rows~~ corresponding source specimens.

~~The environmental-metadata gate reproduced~~ Environmental validation accounts for 274 ~~nonblank source records and verified all 24 exact-cell correction-ledger dispositions against the corresponding raw cells. All retained field temperatures, pressures and relative humidities fell within their declared physical ranges; the unresolved Trip 5 site 40 value of 194% was preserved in the ledger but is absent from both the curated table and the environmental RDF module~~ field observations and checks site identity, temperature, pressure and relative humidity. Values enter the curated environmental dataset only when they satisfy the measurement range and identity criteria. The XRF provenance audit reproduced all 13,390 reported Trips 1–4 cells under the maximum-positive rule and all 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells, and verified that the canonical laboratory input comprises all 725 processed records. An independent chemical-identifier ~~gate checked~~ validation evaluated all 93 instrument channels against the pinned ChEBI ~~release and a release~~ 247 and the checksummed PubChem snapshot of 23 July 2026 (Section 2.9); all rows passed ~~, and audited nulls are retained for the unmapped LE pseudo-analyte~~ the declared rules, with explicit nulls for the aggregate LE channel and unsupported oxides ~~rather than being replaced by lexical matches.~~ The pH ~~reconciliation independently accounts for all~~ validation accounts for 1,168 ~~workbook rows, matches every admitted row to exactly one existing specimen, specimen-assay entries~~ and reproduces the five ~~dispositions reported admission dispositions~~ in Section 32.10. It checks specimen identity for every admitted measurement, including field replicate 2 for all 156 Trip 4 measurements.

4.4 ShEx schema validation

We validated the structural integrity of the knowledge graph using Shape Expressions (ShEx [\[1\]](#)), as [The archived ShEx run used the shapes](#) described in Section 2.1.5. We defined ShEx shapes for all core entity types: sampling sites, soil samples, environmental measurements, DNA extracts, sequencing datasets, quality control records, and taxonomy observations. We bound each shape to its corresponding class instances via a Shape Map.

We implemented the validation using the Apache Jena ShEx library (v4.10.0) invoked from Groovy scripts. We integrated ShEx validation into the automated deployment workflow: each time the workflow regenerates an ABox module, the validation step tests instances of the relevant class against their ShEx shape before loading the data into Virtuoso. The workflow terminates with an error if any instance violates the shape constraints (e.g., a sampling site lacking GPS coordinates or a DNA extract missing a concentration value). This mechanism functions as a regression test, ensuring that modifications to the ETL scripts do not introduce structural inconsistencies. [Its tractable modules passed their assigned shapes.](#) The pH module was validated separately over all 712 measurement processes, values and acidity qualities and all 29 sessions, and the laboratory-control module passed its shape and scientific-invariant checks.

~~The tractable modules passed their assigned ShEx shapes. The ShEx suite also tests rejection, not only conformance: negative fixtures that differ from a valid instance in exactly one respect (a sampling site without coordinates, a value with an invalid datatype. [Four negative fixtures each alter one constraint: missing site coordinates, a an invalid value datatype,](#) reversed value-to-quality direction, [and or](#) an XRF process with neither [an input nor a target](#)) are each rejected by the released shapes, so a passing negative test identifies the constraint that caught it. [input nor target. The released shapes reject all four fixtures.](#)~~ The multi-gigabyte taxonomy ABox remains outside this memory-bounded run; it is instead parsed in full by an independent Turtle parser and scanned by a streaming structural validator, which accounted for 44,528,482 triples in 5,953,690 subject blocks with zero structural violations. This [establishes checks](#) full-file syntax and [generator-specific structural validation,](#) not a claim that the taxonomy ABox passed the unexecuted ShEx shapes.

4.5 Logical consistency

We verified the logical consistency of the knowledge graph using the ELK reasoner, as described in Section 2. We loaded the tractable set of OWL modules into a single OWL ontology and ran ELK to check for unsatisfiable classes and contradictory axioms. The reasoner reported the merged included modules as consistent [the generator’s declared structural relationships. The separate listing validator checks printed Turtle against the modules available for that run. Taxonomy and external reference checks require the corresponding pinned payloads.](#)

We performed this consistency check as part of continuous integration during ontology development. Whenever we modified the TBox or introduced new ABox modules, we re-ran the ELK reasoner to detect regressions. We chose ELK over more expressive reasoners such as HermiT [\[58\]](#) because the size of the knowledge graph made reasoning in the full OWL 2 DL profile computationally infeasible.

4.5 Logical consistency

We implemented the ELK consistency check via the OWL API (v4.5.26) with the ELK OWL API module (v0.4.3) in a Groovy script. A clean rerun on the current tractable modules loaded 781, 293 axioms from 10 distinct ontology IRIs and reported the resulting merged ontology as consistent; the generated ecosystem ELK returned `true` for the tractable scientific-module subset loaded for validation, including the ecosystem, pH and laboratory-control modules. The run excluded the multi-gigabyte taxonomy ABox and large reference modules and did not precompute the class hierarchy, so it establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. ELK is incomplete for universal restrictions, inverse properties and other constructs used by imported modules, and the excluded ABoxes were not covered; likewise, the selected forward-rule program of Section 2.1.4 has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure ⁶⁶.

4.6 Compute-time metrics

. Its execution metadata identify the loaded axioms and import handling; class-hierarchy precomputation was omitted. The ontology uses `xsd:double` literals and constructs outside ELK's supported fragment ⁶⁹; the check therefore applies to the constructs processed by that configuration.

Table 5 summarizes the computational cost of the key deployment steps of the reference Virtuoso deployment. The selected forward-rule expansion (three passes; see Section 2.1.4) runs in approximately 114 minutes. ELK consistency checking over the asserted modules takes under 2 seconds. Competency question latency on the Virtuoso endpoint ranges from 5 ms for simple site look-ups (CQ2) to over 1 second for queries that traverse inverse properties across the entire DNA extraction workflow (CQ5). Validation of the separate finite-rule implementation uses synthetic examples for long hierarchies, transitive cycles, both project property-chain lengths, restriction and intersection classification, and PCR-control ancestry. The tests compare bounded, class-partitioned execution with an unbounded reference closure and check recovery from an interrupted run. Release-specific checks compare the complete control-role inventory with its source, retrieve consequences of the measurement chains and require the inferred graph to be disjoint from the asserted graph. Completion requires the zero-addition pass specified in Section 2.1.4, followed by the explicit clash diagnostics.

The taxonomy composition query (CQ7) exceeds a 5-minute timeout because it joins absolute and relative abundance datasets across 1.4 million observations. This query is not used by the web portal; the portal retrieves taxonomic data per sample rather than across all runs, which completes in sub-second time. Label validation checks that each labelled project entity has exactly one label. The evidence identifies the input modules, their hashes and validator versions (Section 3).

4.6 Usability validation (web portal) Portal access

We developed a web The portal (<https://rubalkhali.science/>) to provide user-friendly access to the data . The application architecture provides an end-to-end validation of

the knowledge graph: the backend dynamically generates SPARQL queries to fetch content directly from the knowledge graph in real time without maintaining a separate relational database. Every page rendered constitutes a functional test of the underlying data. This architecture ensures that the website always reflects the current state of the knowledge graph, and any data integrity problem manifests as a visible error on the portal.

4.7 FAIR compliance

The design decisions described in the preceding sections collectively implement the FAIR Data Principles²⁸. We describe how the resource addresses each principle in turn: retrieves data through SPARQL queries and renders site and specimen views. Validation compares query results with source-derived expected values.

To make the data *findable*, we assigned a globally unique and persistent IRI under the `https://rubalkhali.science/kb/` namespace to every named individual, so that each site, specimen, process, value and dataset is directly and stably addressable rather than identified only by a row in a table. The sequencing data are registered with the European Nucleotide Archive under the umbrella accession PRJEB104209, and the resource is described with rich metadata: the machine-readable manifest indexes every payload file

4.7 Identifiers and reuse metadata

Each named entity has a project-domain IRI. The manifest identifies payload files, and the data dictionary defines every curated field. Because the knowledge graph encodes its own terminology, the metadata and the data share one namespace and can be retrieved together.

To make the data *accessible*, we expose the knowledge graph through a standard SPARQL 1.1 endpoint and a web interface, both of which use open, universally implementable protocols and require no authentication for read access. The raw sequences are retrievable from ENA through their accessions, and the deposited modules can be downloaded and loaded into any RDF store, so access does not depend on the continued operation of our endpoint.

To make the data *interoperable*, we represent all content with established shared vocabularies: SIO as the upper-level framework, ENVO for environmental context, ChEBI for chemical entities, PATO for qualities, UO for units, and NCBI Taxonomy for taxa. The graph uses the standard formal languages OWL 2 DL, RDF/Turtle and SPARQL, and because domain entities are grounded in these widely used ontologies, the resource can be joined with other datasets that use the same vocabularies. The qualified references between modules (for example from a specimen to its site or from a value to its producing process) are themselves asserted as RDF, so the cross-references that give the data their meaning travel with the data.

To make the data *reusable*, we describe the resource with detailed and accurate provenance that records the complete chain from physical sample collection through DNA extraction, library preparation, sequencing and quality control to the derived community profiles, together with the correction and alias ledgers that document every

deterministic repair to the source records. The data are released under CC-BY 4.0, the generation and validation code is open-sourced, specifies curated fields, units and missing-value conventions. RDF provenance links travel with the modules and the release is versioned and archived so that a cited version remains stable. These attributes let a reuser understand the origin and limitations of each observation and integrate the resource into new work with confidence support reuse through SIO, ENVO, ChEBI, PATO, UO and validated taxonomic identifiers ²⁸.

The public endpoint provides read access to the deployed graph. The v3.0.1 Zenodo deposit fixes the released graph, module files and rebuild kit, with checksums, scoped licences and a version-specific DOI. Data Availability identifies the separate sequence-access requirement.

5 Usage notes

The knowledge graph is publicly accessible via the SPARQL endpoint at . Users can submit SPARQL 1.1 queries using standard clients (e.g., curl, Python's SPARQLWrapper, R's SPARQL package). The web portal at <https://rubalkhali.science/> provides a graphical interface for browsing sites, samples, and measurements without requiring SPARQL expertise.

The semantic structure of the knowledge graph enables cross-domain site pages and a SPARQL editor. The endpoint <https://rubalkhali.science/sparql> accepts SPARQL 1.1 queries. For example, researchers investigating the relationship between soil geochemistry and microbial diversity can construct a single query that retrieves calcium concentrations alongside Shannon diversity indices computed from taxon abundances. Executing this query in a relational database would necessitate joining at least four separate tables.

The modular OWL architecture permits extending the knowledge graph with new data types, such as metabolomics or remote sensing data. Users can introduce new classes as subclasses of existing SIO entities and generate corresponding ABox modules. We automate the entire lifecycle from data generation through validation to database loading using the provided deployment workflow (`deploy_onto.sh`).

5.1 Design rationale

We adopted the Semanticscience Integrated Ontology (SIO) as the upper-level framework because it offers a small set of relations (`hasInput`/`hasOutput`, `hasTarget`, `hasMember`, `refersTo`, `hasValue`) that generalize across sampling, measurement, and provenance patterns. Sample collection, XRF analysis, DNA extraction, sequencing, and taxonomic classification are all instantiations of the same a local, low-memory example, load the base, site and XRF modules listed in Table 5 and execute the Site 10 query in Listing 2. Compare its 46 returned observations with the distributed expected identifiers and values. Taxonomic queries also require the taxonomy module and its mapping dependencies. The named-run example retrieves the complete genus profile for `Process--hasInput/hasOutput--Objectidiom`; quantitative results are uniformly represented as Quantity refersTo Quality pairs with explicit units. This uniformity enables cross-domain SPARQL

~~queries that traverse from a climate observation through a sampling site to a taxon abundance without per-domain join logic. Recent work on simplified upper-level ontologies, such as SULO⁷⁰, takes this argument further by strengthening fundamental relations and reducing reliance on specialized predicates; our patterns are compatible with this direction^{ERR16061083}; all-run exports require a locally loaded snapshot and a long-running join. The reproduction instructions accompanying the data provide retrieval, checksum, environment and execution details (Section 3).~~

~~We chose ShEx for ingest validation because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL is an obvious alternative supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time. We deployed Virtuoso as the triple store because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.~~

~~Full OWL 2 DL reasoning over the combined TBox and the ABox is not computationally feasible with HermiT on commodity hardware. We therefore use ELK⁵⁷ to validate the logical consistency of the knowledge graph. For query answering, we rely on a selected forward-rule expansion in Virtuoso, which physically expands the graph by applying subclass, subproperty, inverse, symmetric, equivalence, and transitivity rules; this expansion has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure⁶⁶. Two decades of work on scaling OWL reasoning—materialization-based approaches⁶⁶, query rewriting, modular reasoning, and consequence-based calculi such as ELK itself—provide several alternative points in the design space, and a more expressive reasoner could be applied to a sub-module of the TBox in future work.~~

5.1 Limitations and outstanding challenges

5.1 Choosing observations for reuse

~~Several challenges remain open. First, the assay-aware control screen identifies 351 of 351,472 ASVs using 17 extraction-day blanks with batch links to Use specimen, collection-visit and assay identifiers when joining observations. The primary repeated-campaign frame comprises sites 1–60; the source identifiers 61–64 refer to four named catalogue locations sampled in Trip 1. Several feature profiles share a source specimen identifier. Analyses should group these profiles by their documented specimen and processing relationships.~~

~~The control registry distinguishes assay, processing stage, extraction group and identity evidence. The Trip 5 profiles. Removing these candidates only from the candidate screen uses 17 extraction blanks linked to 217 co-extracted Trip 5 biological profiles removes 2.19% of reads in aggregate, and the unfiltered table remains canonical. Inference is bounded because the design provides only one blank per extraction batch, concentration data are absent, biological profiles and identifies 351 ASVs. The canonical feature table retains these ASVs; the companion sensitivity analysis applies their removal to the linked profiles. Trip 4 has no positive control~~

~~and has an unsequenced blank affecting all or part of 23 sites, and Trip 5 has no six separately denoised extraction blanks and three batches with unsequenced blanks, leaving complete or partial coverage gaps at 28 sites. Concentration measurements exist in the source records; any concentration-based screen requires links between specimens, extractions and concentration measurements. Four Trips 2–3 standards support genus-recovery evaluation. Trips 4–5 have extraction controls but no sequenced 16S positive control. A reuser should use the control module and audit tables to select compatible controls by assay, stage and batch, and should not treat contamination as eliminated. Second, archived-soil pH is available for 712 admitted specimens under the specified positive-standard profiles.~~

~~Archived-soil pH describes specimens measured in CaCl_2 protocol, but coverage is incomplete and non-random and should not be interpreted as an in-situ field map. Third, the XRF sources lack machine-readable unit and non-detect semantics, and Trips 1–4 and Trip 5 use disclosed but different repeat-formula rules; field under the admission rules of Section 2.10. Coverage varies by campaign and compartment. All Trip 4 assays used archived field replicate 2. Their supported comparison with community profiles is at campaign, site and compartment level. The versioned specimen mapping links measurements to the confirmed specimens as described in Methods.~~

~~Field and laboratory XRF values cannot be assumed interchangeable, and their current comparison is observations have separate acquisition provenance and undocumented concentration units. The released values support within-workflow descriptive comparisons; cross-workflow calibration requires documented units, paired aliquots and reference materials. Site 4 field readings include different substrates, so a site-level rather than same-aliquot. Fourth, more complex reasoning—for example over disjunctions arising from ENVO biome classifications or negative axioms about taxon absence—would require either a different reasoner or the extraction of a sub-module amenable to a more expressive calculus. Finally, the resource depends on upstream ontologies (SIO, ChEBI, ENVO, NCBI Taxonomy) that evolve independently; keeping the deployed graph aligned with their releases is an ongoing maintenance task and a known reproducibility risk for any SPARQL query that bottoms out in a specific external IRI summary combines that local heterogeneity with instrument repeatability.~~

~~The PMA table contains three site-compartment groups at two campsites. Each group has one parent collection tube, from which three separate slurries supplied treated/untreated pairs. Descriptive paired summaries retain this preparation structure and compare dye treatment under matched dark and light exposure.~~

5.2 Versioning and supported operations

~~Use the versioned modules and their reference ontologies to reproduce queries. Updating an external ontology requires revalidating identifiers and mappings before regenerating the affected modules and query results.~~

~~KG v3.0.1 serves the asserted graph at <https://rubalkhali.science/graph/asserted/3.0.1>. This graph is the default dataset for portal and SPARQL queries. Explicit FROM, FROM NAMED and SPARQL protocol dataset parameters select the dataset for~~

an individual request. The service evaluates stored assertions directly and rejects unavailable query modes.

5.3 Sustainability and updates

The dataset is versioned and archived on Zenodo, and the deployment workflow is reproducible from the public repository. We intend to extend the resource as additional field expeditions are processed: new trips will be added as additional ABox modules without changes to the upper-level patterns. Upstream ontology updates will be tracked at scheduled intervals; re-materialization is scripted and runs as part of the deployment workflow, so an upstream release can be absorbed by re-running the workflow. Where an upstream change would alter the semantics of an existing query, we will record the affected competency questions in the changelog so that downstream users can audit the impact. The bundled finite-rule implementation supports the separate reasoning workflow described in Section 2.1.4. Publication of its output requires completion and validation of that workflow. Changes to released artifacts require a new semantic version.

A local copy of the deposited graph provides fixed inputs independently of the portal's deployed version. The rebuild kit verifies and loads the 16 distributed modules; generating the scientific modules uses the source data and code described in Section 3. Analyses requiring additional OWL consequences need a reasoner and validation workload suited to those constructs.

6 Data availability

Sequencing data for this project is deposited in the European Nucleotide Archive (ENA) under accession. The project uses ENA umbrella study PRJEB104209; its submitted amplicon run records are associated with PRJEB106069. On 5 October 2026, the anonymous ENA read-run file report for PRJEB106069 returned no public run/file rows. Release of the raw reads or a complete anonymous reviewer download remains required. The source and derived-data package is maintained at <https://github.com/bio-ontology-research-group/empty-quarter-ecology-data>. Its manifests distinguish included files, separately retrieved bulk payloads and unresolved upstream provenance. KG v3.0.1, its original modules and rebuild kit are archived at <https://doi.org/10.5281/zenodo.23134168>⁶⁷. Project-authored RDF, an umbrella study (“The genome of the Rub’ al Khali (Empty Quarter)”, ontology terms, definitions and dataset metadata in this release are licensed under CC BY 4.0. Imported vocabularies retain their respective licences, listed in the release’s third-party attribution file; project software is MIT-licensed. Raw reads and other separately licensed materials retain their own access and licensing conditions. The broader analytical tables and specimen and assay provenance are versioned in the reproducibility repository; the KG deposit contains the files enumerated in its manifest.

7 Code availability

All source code for data processing, ontology engineering, RDF generation, validation, and the web portal is openly available at <https://github.com/bio-ontology-research-group/empty-quarter>. The repository includes Groovy scripts utilizing the OWL API 5 for OWL/RDF generation, Python scripts for data processing and statistical analysis, ShEx shape definitions for structural validation, SPARQL competency queries, React/FastAPI web application source code, Docker Compose configurations for deployment, and the main.nf Nextflow entry point with its pinned software data repository <https://github.com/bio-ontology-research-group/empty-quarter-ecology-data> contains the manuscript sources, RDF generators, metadata and Nextflow workflow. RDF generation uses Groovy scripts with OWL API v5.1.20⁶²; the ELK validation wrapper uses OWL API v4.5.26. The repository supplies execution profiles, workflow tests, a pinned Conda recipe and a hash-locked Python environment.

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Competing interests

The authors declare no competing interests.

References

- [1] Mandaville, J.P.: Plant life in the rub' al-khali (the empty quarter), south-central arabia. Proceedings of the Royal Society of Edinburgh. Section B. Biological Sciences **89**, 147–157 (1986) <https://doi.org/10.1017/s0269727000008988>
- [2] Nelli, N., Francis, D., Sow, M., Fonseca, R., Alkatheeri, A., Bosc, E., Bergametti, G.: The wind-blown sand experiment in the empty quarter desert: Roughness length and saltation characteristics. Earth and Space Science **11**(6), 2024–003512 (2024) <https://doi.org/10.1029/2024EA003512>
- [3] Yasir, M., Azhar, E.I., Khan, I., Bibi, F., Baabdullah, R., Al-Zahrani, I.A., Al-Ghamdi, A.K.: Composition of soil microbiome along elevation gradients in southwestern highlands of Saudi Arabia. BMC Microbiology **15**(1), 65 (2015) <https://doi.org/10.1186/s12866-015-0398-4>
- [4] Makhalanyane, T.P., Valverde, A., Gunnigle, E., Frossard, A., Ramond, J.-B., Cowan, D.A.: Microbial ecology of hot desert edaphic systems. FEMS Microbiology Reviews **39**(2), 203–221 (2015) <https://doi.org/10.1093/femsre/fuu011>

- [5] Fierer, N.: Embracing the unknown: disentangling the complexities of the soil microbiome. *Nature Reviews Microbiology* **15**(10), 579–590 (2017) <https://doi.org/10.1038/nrmicro.2017.87>
- [6] Aslam, S.N., Dumbrell, A.J., Sabir, J.S., Mutwakil, M.H.Z., Baeshen, M.M.N., Abo-Aba, S.E.M., Clark, D.R., Yates, S.A., Baeshen, N.A., Underwood, G.J.C., McGenity, T.J.: Soil compartment is a major determinant of the impact of simulated rainfall on desert microbiota. *Environmental Microbiology* **18**(12), 5048–5062 (2016) <https://doi.org/10.1111/1462-2920.13474>
- [7] Khan, M.A., Khan, S.T.: Microbial communities and their predictive functional profiles in the arid soil of Saudi Arabia. *SOIL* **6**, 513–521 (2020) <https://doi.org/10.5194/soil-6-513-2020>
- [8] Osman, J.R., Zélicourt, A., Bisseling, T., Geurts, R., Hirt, H., DuBow, M.S.: Bacterial rhizosphere biodiversity from several pioneer desert sand plants near Jizan, Saudi Arabia. *The Open Conference Proceedings Journal* **7**(Suppl. 1: M7), 70–79 (2016) <https://doi.org/10.2174/2210289201607020070>
- [9] Eida, A.A., Ziegler, M., Lafi, F.F., Michell, C.T., Voolstra, C.R., Hirt, H., Saad, M.M.: Desert plant bacteria reveal host influence and beneficial plant growth properties. *PLOS ONE* **13**(12), 0208223 (2018) <https://doi.org/10.1371/journal.pone.0208223>
- [10] Maurice, K., Laurent-Webb, L., Dehail, A., Bourceret, A., Boivin, S., Boukcim, H., Selosse, M.-A., Ducousso, M.: Fertility islands, keys to the establishment of plant and microbial diversity in a highly alkaline hot desert. *Journal of Arid Environments* **219**, 105074 (2023) <https://doi.org/10.1016/j.jaridenv.2023.105074>
- [11] Maurice, K., Laurent-Webb, L., Bourceret, A., Boivin, S., Boukcim, H., Selosse, M.-A., Ducousso, M.: Networking the desert plant microbiome, bacterial and fungal symbionts structure and assortativity in co-occurrence networks. *Environmental Microbiome* **19**, 65 (2024) <https://doi.org/10.1186/s40793-024-00610-4>
- [12] Al-Awthan, Y.S., Mir, R., Alatawi, F.A., Alatawi, A.S., Almutairi, F.M., Khafaga, T., Shohdi, W.M., Fakhry, A.M., Alharbi, B.M.: Metagenome analysis identified novel microbial diversity of sandy soils surrounded by natural lakes and artificial water points in King Salman Bin Abdulaziz Royal Natural Reserve, Saudi Arabia. *Life* **14**(12), 1692 (2024) <https://doi.org/10.3390/life14121692>
- [13] Alhasani, F.M., Rajab, M.H., Sulieman, A.M.E., Alanazi, N.A., Aljabry, N., Alwahibi, M.S., Mammdouh, A.: Diversity and ecological drivers of rhizospheric and endophytic microbiomes of desert plants in King Salman Reserve, Saudi Arabia. *Journal of King Saud University – Science* **38**, 16952025 (2026) https://doi.org/10.25259/JKSUS_1695_2025

- [14] Gopi, J., Joydas, T.V., Jayachandran, P.R., Manokaran, S., Mohandas, S.P., Raju, S.C., Hussain, S.A., Abouelresh, M.O., Dixit, S., Rafeeq, M., Asharaf, M., Manikandan, K.P., Hala, S., Pulido, B.: eDNA metabarcoding reveals halophilic and halotolerant bacterial assemblages in the salt flats of the northern Red Sea coast. *Ecological Genetics and Genomics* **38**, 100448 (2026) <https://doi.org/10.1016/j.egg.2026.100448>
- [15] Moussa, T.A.A., Al-Zahrani, H.S., Almaghrabi, O.A., Abdelmoneim, T.S., Fuller, M.P.: Comparative metagenomics approaches to characterize the soil fungal communities of western coastal region, Saudi Arabia. *PLOS ONE* **12**(9), 0185096 (2017) <https://doi.org/10.1371/journal.pone.0185096>
- [16] Murgia, M., Fiamma, M., Barac, A., Deligios, M., Mazzarello, V., Paglietti, B., Cappuccinelli, P., Al-Qahtani, A., Squartini, A., Rubino, S., Al-Ahdal, M.N.: Biodiversity of fungi in hot desert sands. *MicrobiologyOpen* **8**(1), 00595 (2019) <https://doi.org/10.1002/mbo3.595>
- [17] Mani, I., Mikryukov, V., Alkahtani, S., Tedersoo, L.: Desert mycobiome of Saudi Arabia is driven by vegetation patterns. *MycoKeys* **130**, 71–100 (2026) <https://doi.org/10.3897/mycokeys.130.176937>
- [18] Blanco-Sacristán, J., Nelson, C., Corrochano-Monsalve, M., Maestre, F.T.: Biological soil crusts in the Arabian Peninsula: Ecological functions, current knowledge and research gaps. *Cambridge Prisms: Drylands* **2**, 13 (2025) <https://doi.org/10.1017/dry.2025.10007>
- [19] McKay, C.P., Rask, J.C., Detweiler, A.M., Bebout, B.M., Everroad, R.C., Lee, J.Z., Chanton, J.P., Mayer, M.H., Caraballo, A.A.L., Kapili, B., Al-Awar, M., Al-Farraj, A.: An unusual inverted saline microbial mat community in an interdune sabkha in the Rub' al-Khali (the Empty Quarter), United Arab Emirates. *PLOS ONE* **11**(3), 0150342 (2016) <https://doi.org/10.1371/journal.pone.0150342>
- [20] Mousa, W.K., Abu-Izneid, T., Salah-Tantawy, A.: High-throughput sequencing reveals the structure and metabolic resilience of desert microbiome confronting climate change. *Frontiers in Plant Science* **15**, 1294173 (2024) <https://doi.org/10.3389/fpls.2024.1294173>
- [21] Sharma, A., Kosasih, E., Zhang, J., Brintrup, A., Calinescu, A.: Digital twins: State of the art theory and practice, challenges, and open research questions. *Journal of Industrial Information Integration* **30**, 100383 (2022) <https://doi.org/10.1016/j.jii.2022.100383>
- [22] Bauer, P., Stevens, B., Hazeleger, W.: A digital twin of Earth for the green transition. *Nature Climate Change* **11**(2), 80–83 (2021) <https://doi.org/10.1038/s41558-021-00986-y>

- [23] Khan, T., Koning, K., Endresen, D., Chala, D., Kusch, E.: TwinEco: A unified framework for dynamic data-driven digital twins in ecology. *Ecological Informatics* **91**, 103407 (2025) <https://doi.org/10.1016/j.ecoinf.2025.103407>
- [24] Iglesias, F., Ros, F., Hoang Vy Thuy, L., Gourcy, L., Moquet, J.-S., Daële, V., Dupraz, S.: A conceptual architecture for AI-assisted digital twins in natural resource management. *Ecological Informatics* **94**, 103635 (2026) <https://doi.org/10.1016/j.ecoinf.2026.103635>
- [25] Ovaskainen, O., Winter, S., Tikhonov, G., Lauha, P., Lehtiö, A., Nokelainen, O., Abrego, N., Aroluoma, A., Harrison, J.P., Heikkinen, M., *et al.*: A digital twin for real-time biodiversity forecasting with citizen science data. *Nature Ecology & Evolution* **10**(3), 481–495 (2026) <https://doi.org/10.1038/s41559-025-02966-3>
- [26] Islam, S., Koivula, H., Andrew, C., Lopez Gordillo, J., Weiland, C., Schigel, D., Endresen, D., Arvanitidis, C., Chadwick, E., Soiland-Reyes, S.: FAIR digital twins for biodiversity: enabling data, model, and workflow integration. *npj Biodiversity* **5**(1), 5 (2026) <https://doi.org/10.1038/s44185-025-00116-3>
- [27] Marshall-Colon, A., Long, S.P., Allen, D.K., Allen, G., Beard, D.A., Benes, B., Caemmerer, S., Christensen, A.J., Cox, D.J., Hart, J.C., *et al.*: Towards a digital twin for plant biology. *Trends in Plant Science* **22**(12), 1024–1033 (2017)
- [28] Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L.B., Bourne, P.E., *et al.*: The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* **3**(1), 160018 (2016)
- [29] Pesant, S., Not, F., Picheral, M., Kandels-Lewis, S., Le Bescot, N., Gorsky, G., Iudicone, D., Karsenti, E., Speich, S., Troublé, R., Dimier, C., Searson, S., Tara Oceans Consortium Coordinators: Open science resources for the discovery and analysis of Tara Oceans data. *Scientific Data* **2**, 150023 (2015) <https://doi.org/10.1038/sdata.2015.23>
- [30] Yilmaz, P., Kottmann, R., Field, D., Knight, R., Cole, J.R., Amaral-Zettler, L., Gilbert, J.A., *et al.*: Minimum information about a marker gene sequence (MIMARKS) and minimum information about any (x) sequence (MIxS) specifications. *Nature Biotechnology* **29**(5), 415–420 (2011) <https://doi.org/10.1038/nbt.1823>
- [31] Kelliher, J.M., Mirzayi, C., Bordenstein, S.R., Oliver, A., Kellogg, C.A., Hatcher, E.L., Berg, M., Baldrian, P., Aljumaah, M., *et al.*: STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies. *Nature Microbiology* **10**(12), 3059–3068 (2025) <https://doi.org/10.1038/s41564-025-02186-2>

- [32] National Microbiome Data Collaborative: National Microbiome Data Collaborative: Data Standards. <https://microbiomedata.org/data-standards/>. Accessed 23 July 2026 (2026)
- [33] Santangelo, B.E., Hegde, H., Caufield, J.H., Reese, J., Kliegr, T., Hunter, L.E., Lozupone, C.A., Mungall, C.J., Joachimiak, M.P.: KG-Microbe – building modular and scalable knowledge graphs for microbiome and microbial sciences. *GigaScience*, 077 (2026) <https://doi.org/10.1093/gigascience/giag077>
- [34] Ma, C., Liu, S., Won, S., Koslicki, D.: MetagenomicKG: a knowledge graph for metagenomic applications. *Bioinformatics* **42**(7), 421 (2026) <https://doi.org/10.1093/bioinformatics/btag421>
- [35] W3C OWL Working Group: OWL 2 Web Ontology Language Document Overview (Second Edition). W3c recommendation, W3C (December 2012). <https://www.w3.org/TR/owl2-overview/>
- [36] Cyganiak, R., Wood, D., Lanthaler, M.: RDF 1.1 Concepts and Abstract Syntax. W3c recommendation, W3C (February 2014). <https://www.w3.org/TR/rdf11-concepts/>
- [37] Open-Meteo: Open-Meteo: free weather API. <https://open-meteo.com/> (2024)
- [38] Dumontier, M., Baker, C.J., Baran, J., Callahan, A., Chepelev, L., Cruz-Toledo, J., Del Rio, N.R., Duck, G., Furlong, L.I., Keath, N., *et al.*: The SemanticScience Integrated Ontology (SIO) for biomedical research and knowledge discovery. *Journal of Biomedical Semantics* **5**(1), 14 (2014) <https://doi.org/10.1186/2041-1480-5-14>
- [39] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl.1), 344–350 (2008) <https://doi.org/10.1093/nar/gkm791>
- [40] Buttigieg, P.L., Morrison, N., Smith, B., Mungall, C.J., Lewis, S.E.: The environment ontology: contextualising biological and biomedical entities. *Journal of Biomedical Semantics* **4**(1), 43 (2013)
- [41] Schoch, C.L., Ciufo, S., Domrachev, M., Hotton, C.L., Kannan, S., Khovanskaya, R., Leipe, D., Mcveigh, R., O’Neill, K., Robbertse, B., *et al.*: NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database* **2020**, 062 (2020) <https://doi.org/10.1093/database/baaa062>
- [42] Tawfiq, R., Abdelhakim, M., Alajel, S.M., Alarawi, M., Aldakhil, H., Derouiche, A., Drautz-Moses, D.I., Dumontier, M., Grünberg, R., Lopez-Velazquez, A., Martinez Arbas, S., Niu, K., Sivakumar, K., Wang, T., Zhao, X., Zubair, J., Rueping, M., Hoehndorf, R.: Landscape-scale bacterial biogeography across the

Rub' al-Khali. Companion ecology manuscript in preparation (2026)

- [43] Klindworth, A., Pruesse, E., Schweer, T., Peplies, J., Quast, C., Horn, M., Glöckner, F.O.: Evaluation of general 16s ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. *Nucleic Acids Research* **41**(1), 1 (2013) <https://doi.org/10.1093/nar/gks808>
- [44] Illumina: 16s metagenomic sequencing library preparation: Preparing 16s ribosomal rna gene amplicons for the illumina miseq system (2013). Available at https://support.illumina.com/documents/documentation/chemistry_documentation/16s/16s-metagenomic-library-prep-guide-15044223-b.pdf; accessed 27 August 2026
- [45] Ewels, P.A., Peltzer, A., Fillinger, S., Patel, H., Alneberg, J., Wilm, A., Garcia, M.U., Di Tommaso, P., Nahnsen, S.: The nf-core framework for community-curated bioinformatics pipelines. *Nature Biotechnology* **38**(3), 276–278 (2020) <https://doi.org/10.1038/s41587-020-0439-x>
- [46] Straub, D., Blackwell, N., Langarica-Fuentes, A., Peltzer, A., Nahnsen, S., Klein-dienst, S.: Interpretations of environmental microbial community studies are biased by the selected 16s rna (gene) amplicon sequencing pipeline. *Frontiers in Microbiology* **11**, 550420 (2020)
- [47] Callahan, B.J., McMurdie, P.J., Rosen, M.J., Han, A.W., Johnson, A.J.A., Holmes, S.P.: DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods* **13**(7), 581–583 (2016) <https://doi.org/10.1038/nmeth.3869>
- [48] Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glöckner, F.O.: The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Research* **41**(D1), 590–596 (2013) <https://doi.org/10.1093/nar/gks1219>
- [49] Fierer, N., Leung, P.M., Lappan, R., Eisenhofer, R., Ricci, F., Holland, S.I., *et al.*: Guidelines for preventing and reporting contamination in low-biomass microbiome studies. *Nature Microbiology* **10**(7), 1570–1580 (2025) <https://doi.org/10.1038/s41564-025-02035-2>
- [50] Douglas, G.M., Maffei, V.J., Zaneveld, J.R., Yurgel, S.N., Brown, J.R., Taylor, C.M., Huttenhower, C., Langille, M.G.I.: PICRUSt2 for prediction of metagenome functions. *Nature Biotechnology* **38**(6), 685–688 (2020) <https://doi.org/10.1038/s41587-020-0548-6>
- [51] Balvočiūtė, M., Huson, D.H.: Silva, rdp, greengenes, ncbi and ott—how do these taxonomies compare? *BMC genomics* **18**(Suppl 2), 114 (2017)

- [52] Parks, D.H., Chuvochina, M., Rinke, C., Mussig, A.J., Chaumeil, P.-A., Hugenholtz, P.: GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Research* **50**(D1), 785–794 (2022) <https://doi.org/10.1093/nar/gkab776>
- [53] Aroney, S.T.N., Newell, R.J.P., Nissen, J.N., Camargo, A.P., Tyson, G.W., Woodcroft, B.J.: CoverM: read alignment statistics for metagenomics. *Bioinformatics* **41**(4), 147 (2025) <https://doi.org/10.1093/bioinformatics/btaf147>
- [54] Cantalapiedra, C.P., Hernández-Plaza, A., Letunic, I., Bork, P., Huerta-Cepas, J.: eggNOG-mapper v2: functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Molecular Biology and Evolution* **38**(12), 5825–5829 (2021) <https://doi.org/10.1093/molbev/msab293>
- [55] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)
- [56] Prud’hommeaux, E., Labra Gayo, J.E., Solbrig, H.: Shape Expressions: an RDF validation and transformation language. In: *Proceedings of the 10th International Conference on Semantic Systems (SEM ’14)*, pp. 32–40. ACM, Leipzig, Germany (2014). <https://doi.org/10.1145/2660517.2660523>
- [57] Kazakov, Y., Krötzsch, M., Simančík, F.: The incredible ELK: From polynomial procedures to efficient reasoning with $\mathcal{EL}$ ontologies. *Journal of Automated Reasoning* **53**(1), 1–61 (2014) <https://doi.org/10.1007/s10817-013-9296-3>
- [58] Glimm, B., Horrocks, I., Motik, B., Stoilos, G., Wang, Z.: HermiT: an OWL 2 reasoner. *Journal of Automated Reasoning* **53**(3), 245–269 (2014)
- [59] Baader, F., Calvanese, D., McGuinness, D.L., Nardi, D., Patel-Schneider, P.F.: *The Description Logic Handbook: Theory, Implementation, and Applications*, 2nd edn. Cambridge University Press, Cambridge (2007). <https://doi.org/10.1017/CBO9780511711787>
- [60] Gkoutos, G.V., Green, E.C., Mallon, A.-M., Hancock, J.M., Davidson, D.: The Phenotype And Trait Ontology (PATO): a standard for cross-species phenotype data integration. *Mammalian Genome* **16**, 866–879 (2005)
- [61] Gkoutos, G.V., Schofield, P.N., Hoehndorf, R.: The Units Ontology: a tool for integrating units of measurement in science. *Database* **2012**, 033 (2012)
- [62] Horridge, M., Bechhofer, S.: The OWL API: a Java API for OWL ontologies. *Semantic Web* **2**(1), 11–21 (2011)

- [63] Cuenca Grau, B., Horrocks, I., Kazakov, Y., Sattler, U.: Modular reuse of ontologies: theory and practice. *Journal of Artificial Intelligence Research* **31**, 273–318 (2008) <https://doi.org/10.1613/jair.2375>
- [64] OpenLink Software: OpenLink Virtuoso Universal Server. <https://virtuoso.openlinksw.com/> (2024)
- [65] Grau, B.C., Horrocks, I., Kazakov, Y., Sattler, U.: A survey on ontology modularization. *Theoretical Computer Science* **391**(3), 198–233 (2008)
- [66] Horst, H.J.: Completeness, decidability and complexity of entailment for rdf schema and a semantic extension involving the owl vocabulary. *Journal of Web Semantics* **3**(2–3), 79–115 (2005) <https://doi.org/10.1016/j.websem.2005.06.001>
- [67] Tawfiq, R., et al.: The microbial diversity of Rub’ al-Khali (Empty Quarter). Zenodo (2026). <https://doi.org/10.5281/zenodo.23134168> . <https://doi.org/10.5281/zenodo.23134168>
- [68] Harris, S., Seaborne, A.: SPARQL 1.1 Query Language. W3c recommendation, W3C (March 2013). <https://www.w3.org/TR/sparql11-query/>
- [69] Motik, B., Cuenca Grau, B., Horrocks, I., Wu, Z., Fokoue, A., Lutz, C.: OWL 2 Web Ontology Language Profiles (Second Edition). W3c recommendation, W3C (December 2012). <https://www.w3.org/TR/owl2-profiles/>
- [70] Dumontier, M., Çelebi, R., Gilani, K., Zegher, I., Serafimova, K., Martínez Costa, C., Schulz, S.: SULO – a simplified upper-level ontology. In: *Proceedings of the Joint Ontology Workshops (JOWO) – Episode XI: The Sicilian Summer Under the Etna, Co-located with the 15th International Conference on Formal Ontology in Information Systems (FOIS 2025)*, Catania, Italy (2025). https://github.com/AIDAVA-DEV/sulo/blob/main/docs/papers/FOUST2025/SULO_FOUST2025.pdf